

Lean – Six Sigma – Capability / SPC

UGain Black Belt in Lean – module 3: Geavanceerde Technieken in Lean – Six Sigma
Onderwerp: 2.5, lean toolbox referentie: 13

Dries Grymonprez

dries.grymonprez@arcelormittal.com



Who is who?

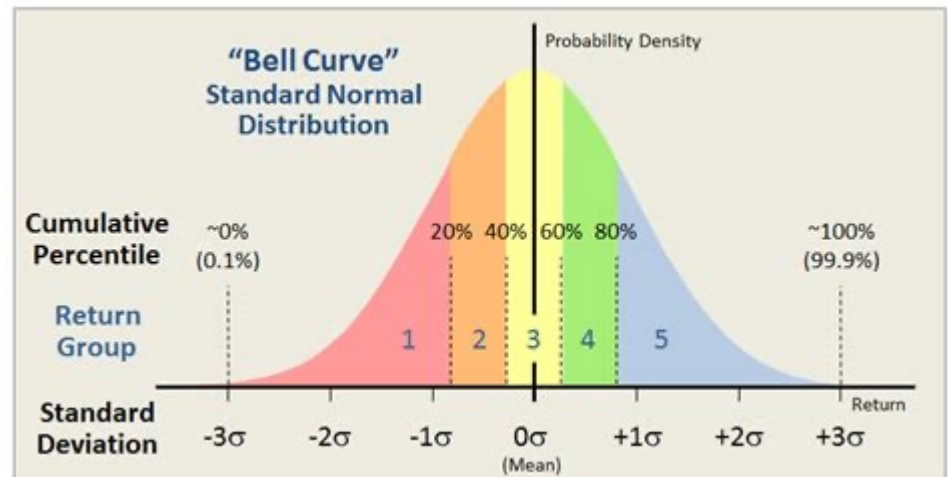
- Dries Grymonprez, ArcelorMittal Gent, CAPEX mgr / decarb
Six Sigma Black Belt, 2005, dries.grymonprez@arcelormittal.com
- ArcelorMittal Belgium (www.arcelormittal.com/gent) – Steel Industry
Key messages & Facts & Key figures – see company film
<https://www.youtube.com/watch?v=kLvIlvK-yVA>
- Data = basis of everything
(incorporation of data analysis & statistics in company culture)
- Maximal automation, modelling, already in 1970s
- Top performing people (5 % of working time invested in training)
- Each trainee can presents her- or himself + expectations
- Exam

Wat zullen we behandelen vandaag?

- Data
 - Basisstatistiek: normale verdeling: spreiding (minimaliseren)
 - Stabiliteit
 - Capabiliteit → vs. klanteisen (C_p , C_{pk})
 - Control: SPC: statistische proces controle: regelkaarten

Wat zullen we behandelen vandaag?

- DATA als basis
- STATISTIEK
Normale verdeling – statistische technieken voor inzicht en om problemen op te lossen



Agenda

- 13u30-15u30: deel 1 Process Capability
- 15u30-15u45: pauze
- 15u45-17u45: deel 2 Statistical Process Control (charts)
- 17u45-18u15: pauze (broodjes)
- 18u15-20u15: deel 3 Exercises

Leerdoelstellingen

- Wat is capabiliteit van een proces en kunnen toepassen
- Statistische proces controle:
 - Concept en toepasbaarheid
 - Hoe stellen we een eenvoudige regelkaart op
 - Wat is out of control en hoe sturen we bij?
- Oefeningen + interactiviteit
- **Algemene doelstelling:**
belang van data verzamelen/analyse kennen & kunnen toepassen

Agenda

- 13u30-15u30: deel 1 Intro & Process Capability
- 15u30-15u45: pauze
- 15u45-17u45: deel 2 Statistical Process Control (charts)
- 17u45-18u15: pauze (broodjes)
- 18u15-20u15: deel 3 Exercises

Waarde creëren dankzij slimme data

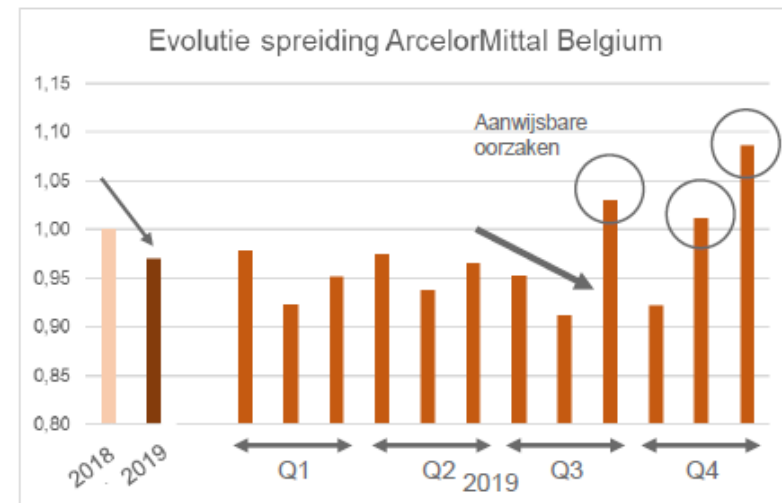
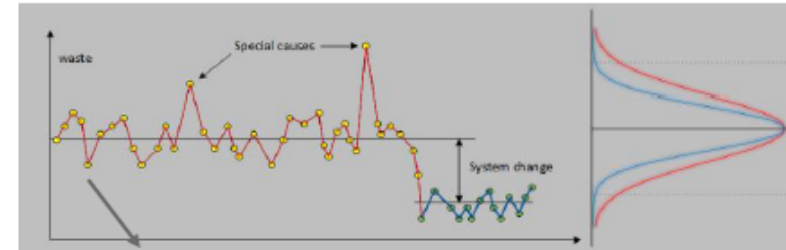
Wij willen **statistisch “denken”** in onze ‘business’ en industriële processen:

Statistisch “denken” = beschikbaarheid van data + toegepaste statistiek (PyDex) + “mindset”

- ➔ spreiding meten = inzicht verwerven
- ➔ spreiding verminderen = verliezen bestrijden

Voordelen:

- Processen verbeteren
- Evolueren van een reactieve naar een preventieve modus
- Data naar de mensen brengen via gebruik van mobiele toepassingen
- Minder verliezen, lagere kosten en snellere verbetering van de prestaties
- Minder energieverlies, meer groene tonnen

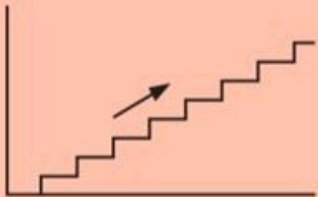


Methodologies landscape

Kaizen

- Continu verbeteren
- Snel kleine verbeteringen
- Eenvoud
- Met en op de werkvloer
- Productie en distributie
- 5S-aanpak

Verbeteren in kleine stappen

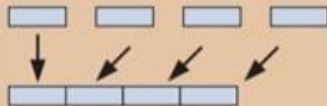


© Hessel Visser

Lean

- Elimineren verspilling zoals voorraden
- Standaardiseren
- Korte omsteltijden
- Creëren flexibele flow
- Kanban aansturing
- Productie en distributie
- TPM
- Brede aanpak

Weghalen van verspilling uit de keten



Agile

- Snelheid
- Reactie
- Flexibiliteit
- Short time to market
- Wispelturige klanten
- Onvoorspelbaar gedrag

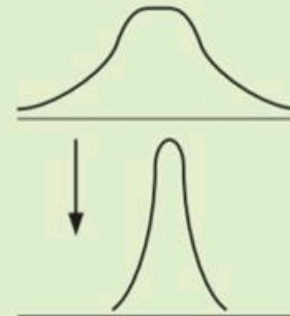
Alert reageren op verandering



Six Sigma

- DMAIC-proces
- Complexere problemen
- Statistiek
- Variatieanalyse
- Reductie variatie
- Experimenteren
- Bijna overal toepasbaar
- Diepteaanpak

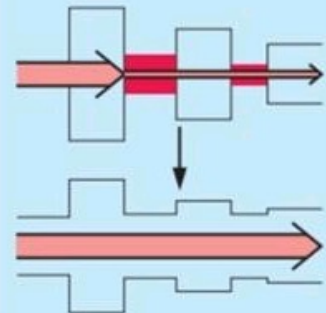
Elimineren van variatie



TOC

- Bottleneck ontdekken
- Vijf stappen om bottleneck te elimineren
- Oorzaak- en gevolganalyse
- Drum-buffer-rope
- Kritieke pad ontdekken
- Continu herplannen

Maximaliseren van de output



What is Six Sigma?*

Wat is Six Sigma?

Je kunt Six Sigma op diverse manieren definiëren:

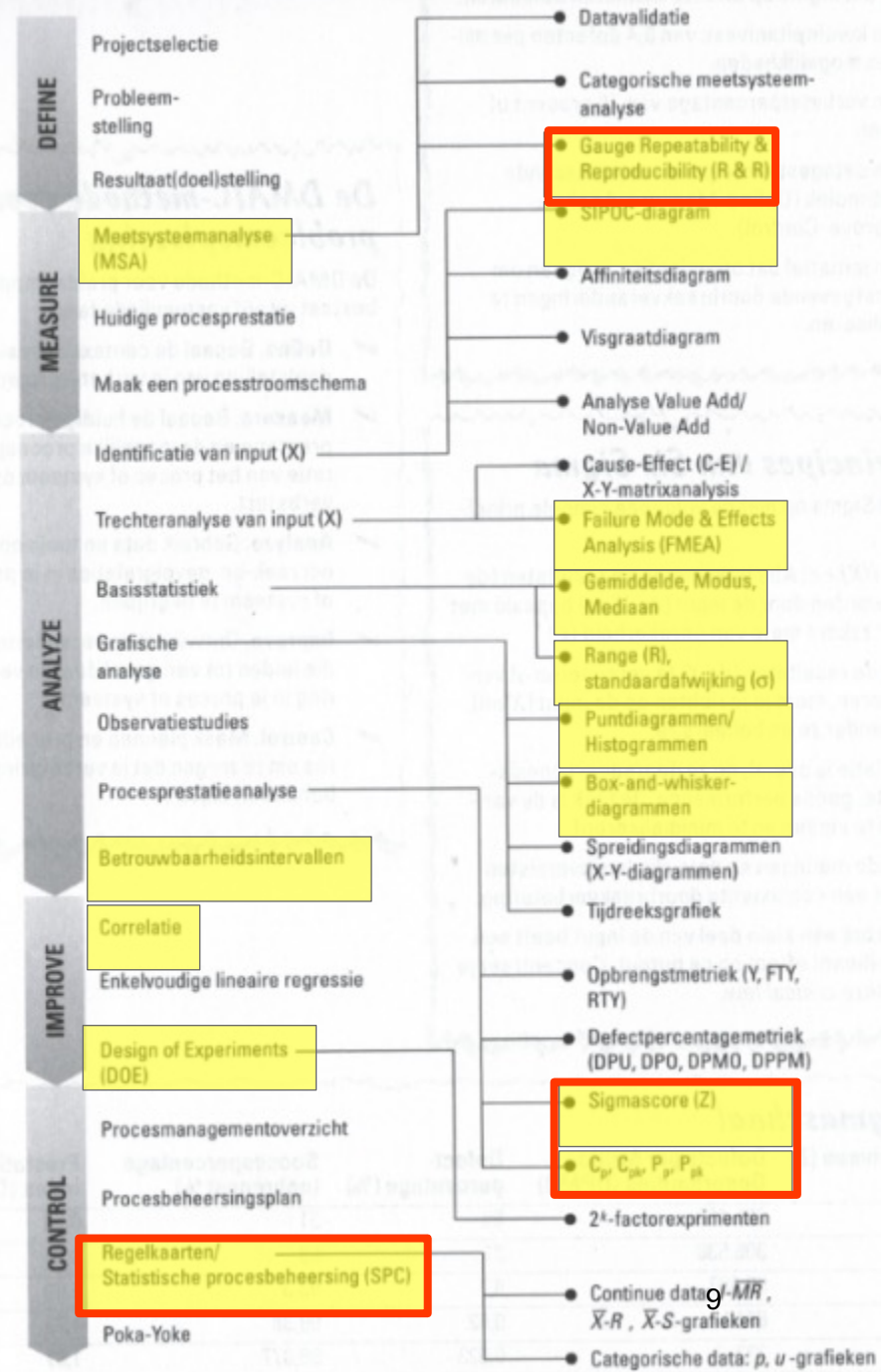
- ✓ Een kwaliteitsniveau van 3,4 defecten per miljoen mogelijkheden.
- ✓ Een verbeterpercentage van 70 procent of meer.
- ✓ Een datagestuurde probleemoplossende methodiek (Define-Measure-Analyze-Improve-Control).
- ✓ Een initiatief dat organisaties aangaan om winstgevende doorbraakveranderingen te realiseren.

De DMAIC-methode voor probleemoplossing

De DMAIC-methode voor probleemoplossing bestaat uit vijf vastomlijnde fasen.

- ✓ **Define.** Bepaal de context en resultaat-doelstelling van je verbeterproject.
- ✓ **Measure.** Bepaal de huidige procesprestatie en de mogelijke procesprestatie van het proces of systeem dat je verbetert.
- ✓ **Analyze.** Gebruik data en tools om de oorzaak-en-gevolgrelaties in je proces of systeem te begrijpen.
- ✓ **Improve.** Ontwikkel de veranderingen die leiden tot een gevalideerde verbetering in je proces of systeem.
- ✓ **Control.** Maak plannen en procedures om te zorgen dat je verbeteringen behouden blijven.

Het Six Sigma-landschap van tools en methoden



Six Sigma scientific - statistical point of view

- **Dispersion** everywhere
(entropy – second law of thermodynamics) – $dS/dt \geq 0$
Dispersion is expressed in variance / standard deviation (sigma)
- Control dispersion → zero defects
- Statistics – descriptive/analysis
- **Standard normal distribution** – also other distributions
- **Nothing is sure** – always a chance that a hypothesis is wrong
E.g., weather forecast

Data collection

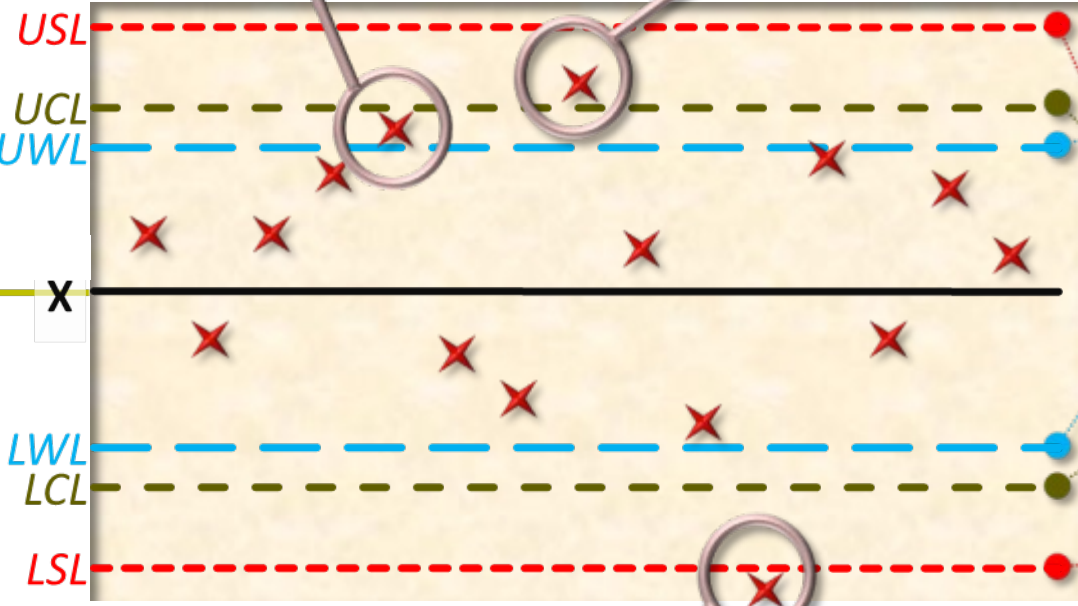
- As basis: observations & measurements to collect data (simple means, lab, on-line) with steps:
 1. Define well what you want to measure (what matters, independent variables)
 2. Choose data collection method: experiments, on-line, but also: surveys, interviews, ...
 3. Plan your data collection procedures (how will you measure, how accurate, what's your sampling plan?, standardization of measurements)
 4. Data collection
- Data storage: on paper, but also of course digitalization, store in databases, for further analyses/studies

Sample size determination

- Is always trade-off between accuracy of modelling & cost
- Larger samples may provide more accurate information, but may also not provide additional accuracy or may even be dangerous: long-term effects may be included
- General rule: sample at least 20, preferably 30 measurements
- Sample size determination is a study on its own, not an objective of this course.
- More info: https://en.wikipedia.org/wiki/Sample_size_determination

Upper limits
→
Lower limits
←

Increase attention?
Out of control?



Warning limits:
Does the process get proper attention or not?

Control limits:
Is the process in control or not?

Specification limits:
Shall we accept or reject the item?

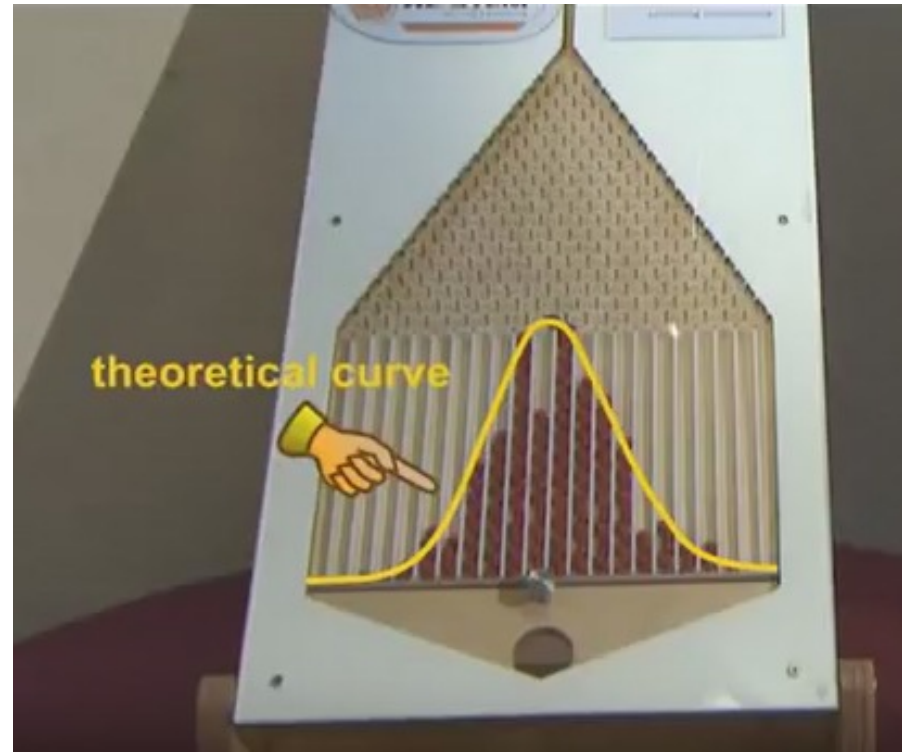
✕ Measurement

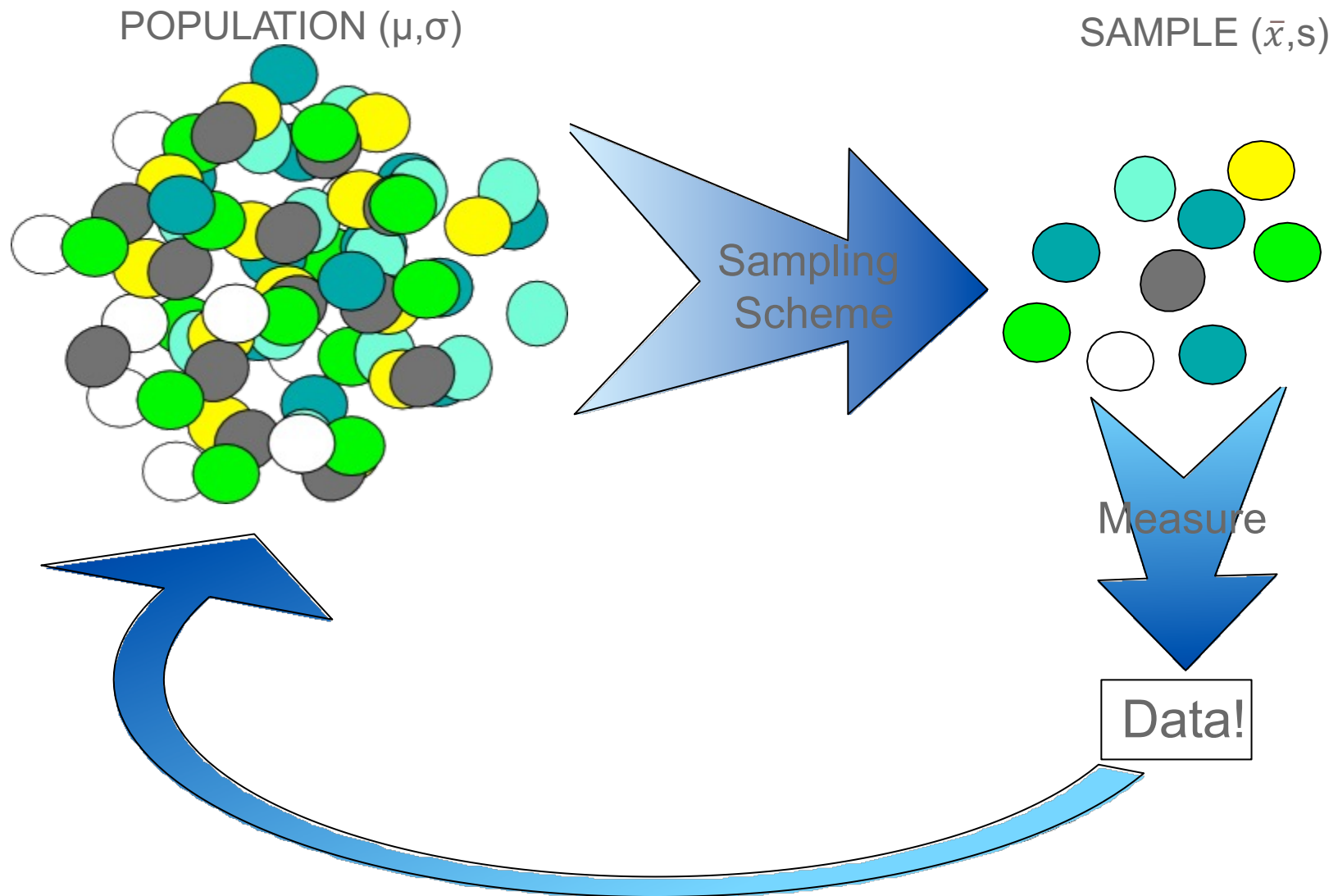
Reject?

Galton Board

- Galton Board

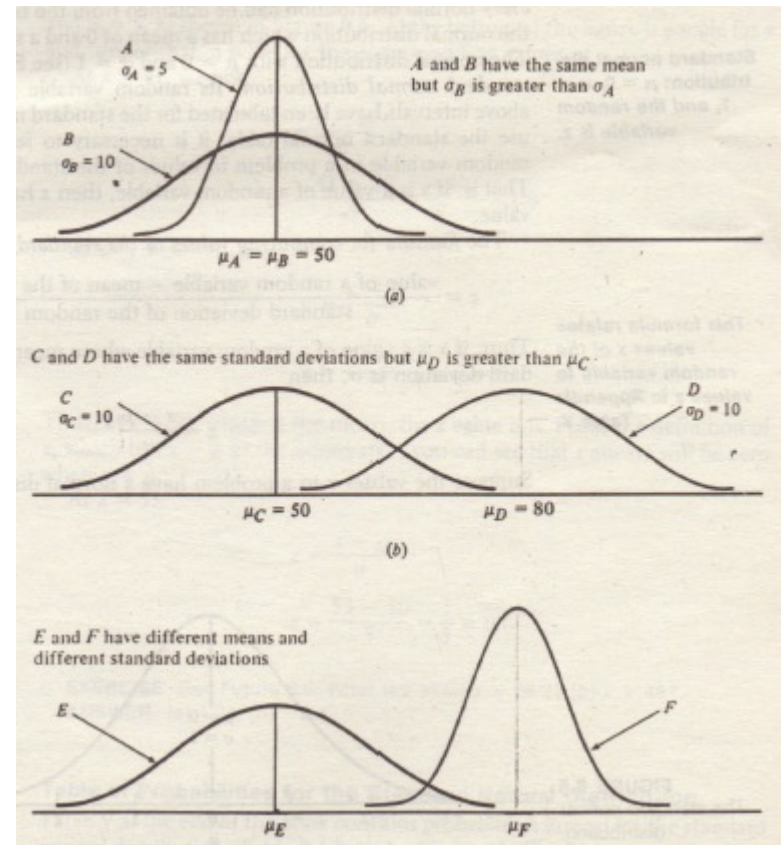
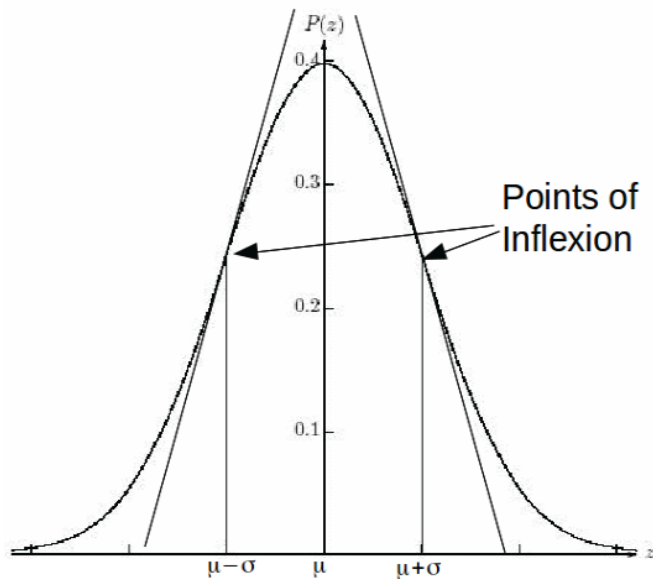
<https://www.youtube.com/watch?v=6YDHBFIvls>



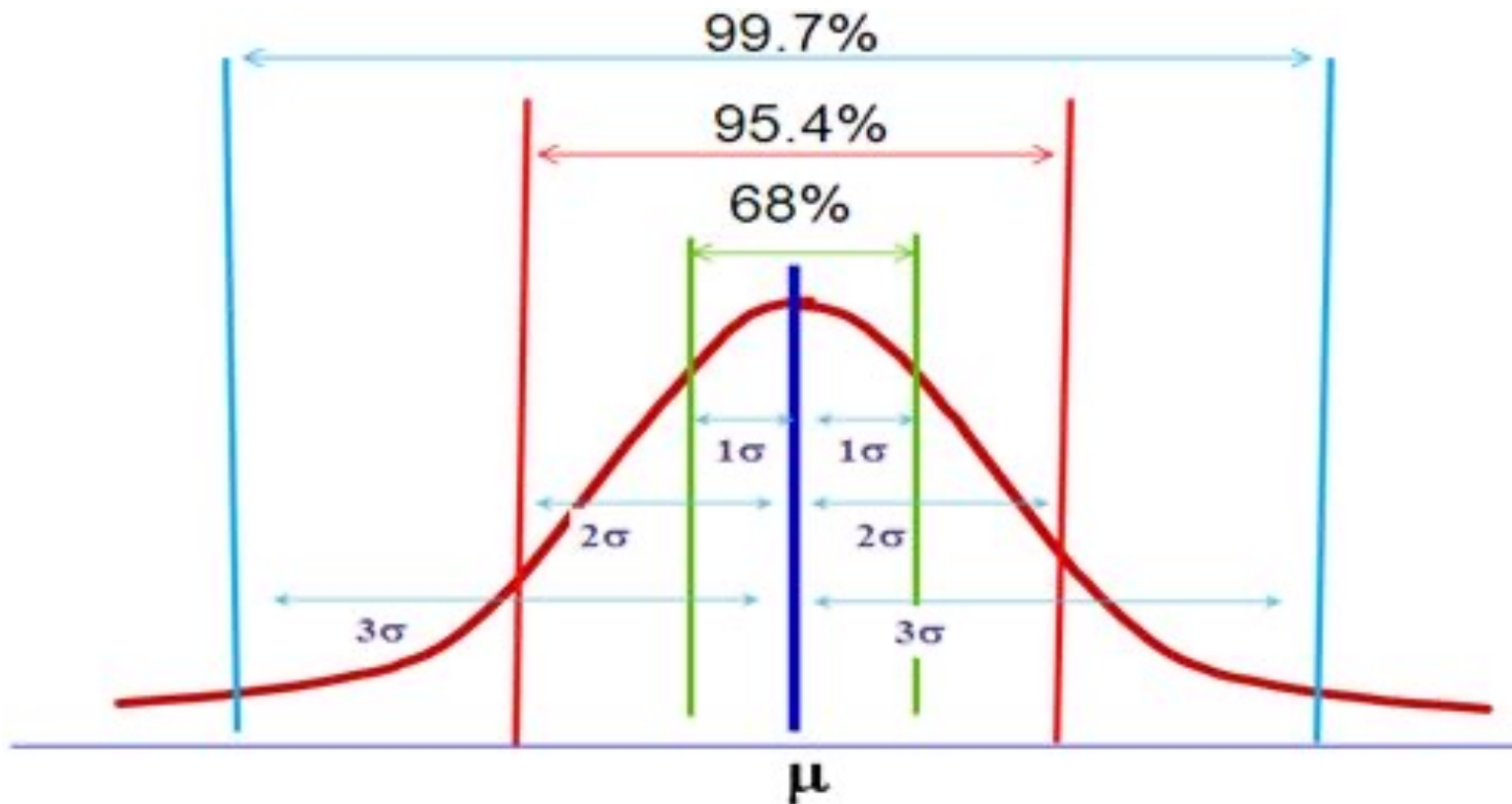


Normal distribution

- Probability distribution (symetric)
- Surface under curve = 100 %
- Kurtosis ('peak'ness)
- Definition of sigma = inflexion point



- Hoeveel nog buiten 3 Sigma? 0,30 % totaal, dus 0,15 % links/rechts



(Standaard)normale verdeling

- Laat toe om steekproeven uit verschillende populaties op een zinvolle manier met elkaar te vergelijken én gemakkelijker te rekenen
- $X \sim N(\mu, \sigma)$ met N = beste schatting
 - $E(X) = \mu$
 - $\sigma \quad \sigma(X) = \sigma$
 - Belangrijke kengetallen :
 - $P[\mu - \sigma \leq X \leq \mu + \sigma] \approx 68 \%$
 - $P[\mu - 2\sigma \leq X \leq \mu + 2\sigma] \approx 95 \%$
 - $P[\mu - 3\sigma \leq X \leq \mu + 3\sigma] \approx 99.7 \%$
- Standaard normale Z belangrijk voor praktijk:
 - $E(Z) = 0$
 - $\sigma \quad \sigma(Z) = 1$

$$Z = \frac{X - \mu}{\sigma}$$

Calculating Process Capability

Discrete Data

Yield and Defect Count

Yield and Defect Counts are related measures

Yield represents the percent of good products or services (measured in %)

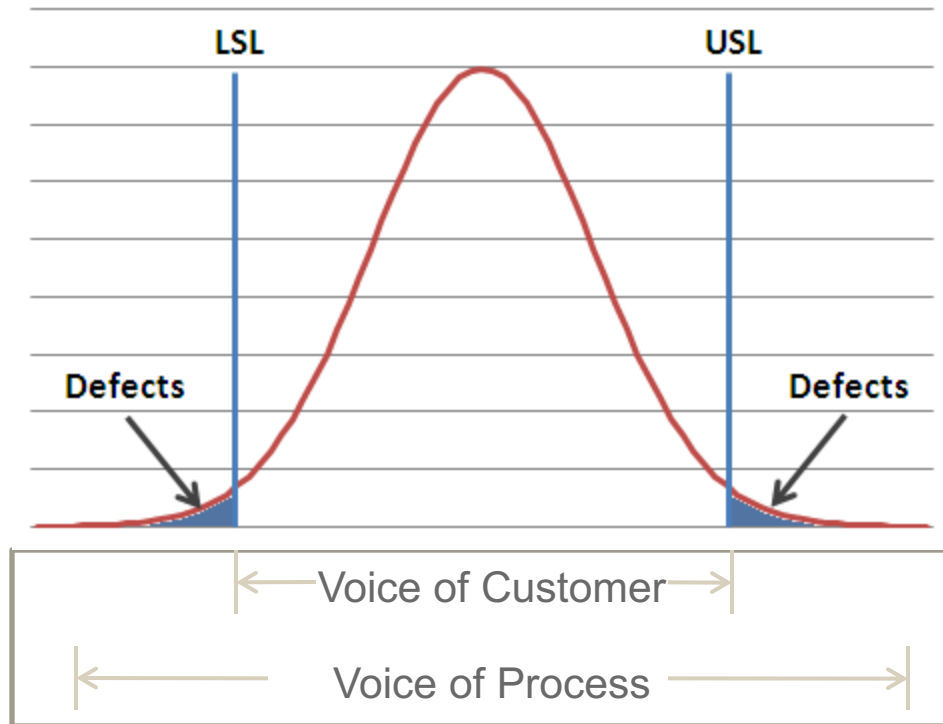
Defect Counts monitor the number of times things go wrong [measured in Defects Per Opportunity (DPO) or Defects Per Million Opportunities (DPMO)]

PPM = parts per million
(+ % to PPM: factor 10.000)

Calculations
$DPO = \frac{D}{N \cdot O}$
$DPMO = 1,000,000 \cdot \frac{D}{N \cdot O}$
$Yield = 1 - \frac{D}{N \cdot O}$

Key: D = Number of defects
N = Number of units
O = Opportunities per unit

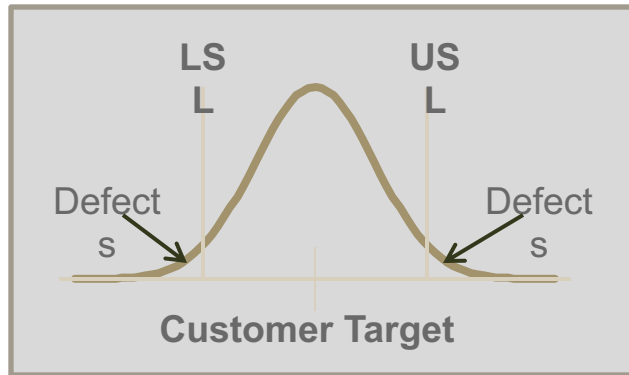
VOC vs. VOP



Sigma Capability	Defects per Million Opportunities	% Yield
2	308,537	69.15%
3	66,807	93.32%
4	6,210	99.38%
5	233	99.98%
6	3.4	99.99966%

The Voice of the Process is
independent of the Voice of the
Customer

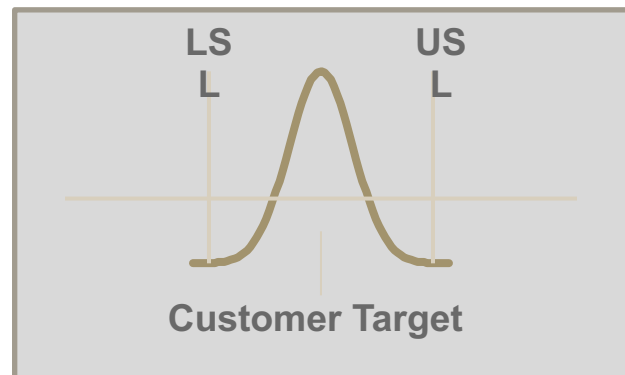
Goals of Lean Six Sigma



Prevent Defects by
Reducing Variation

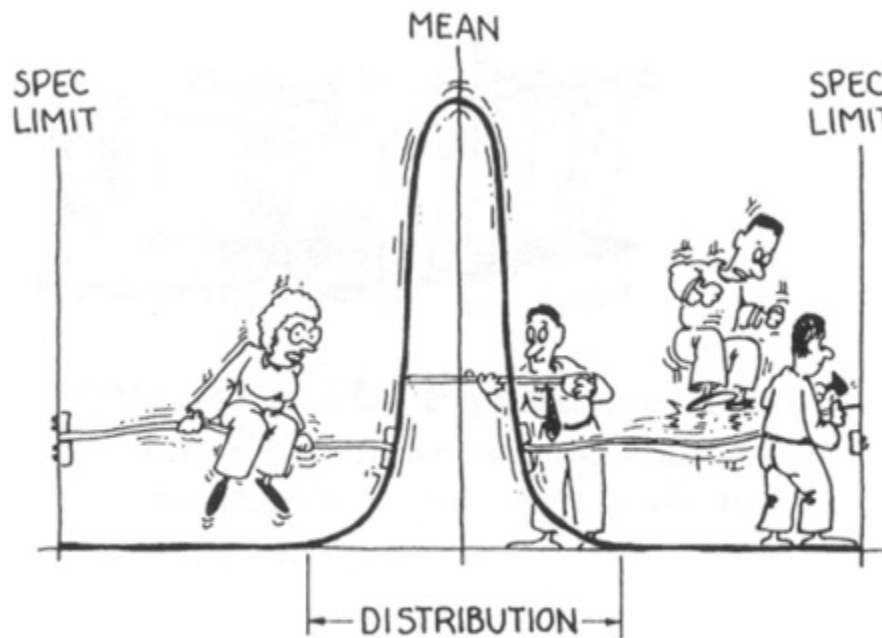


Prevent Defects by
Centering Process



Meet Customer
Requirements

Six Sigma = team work

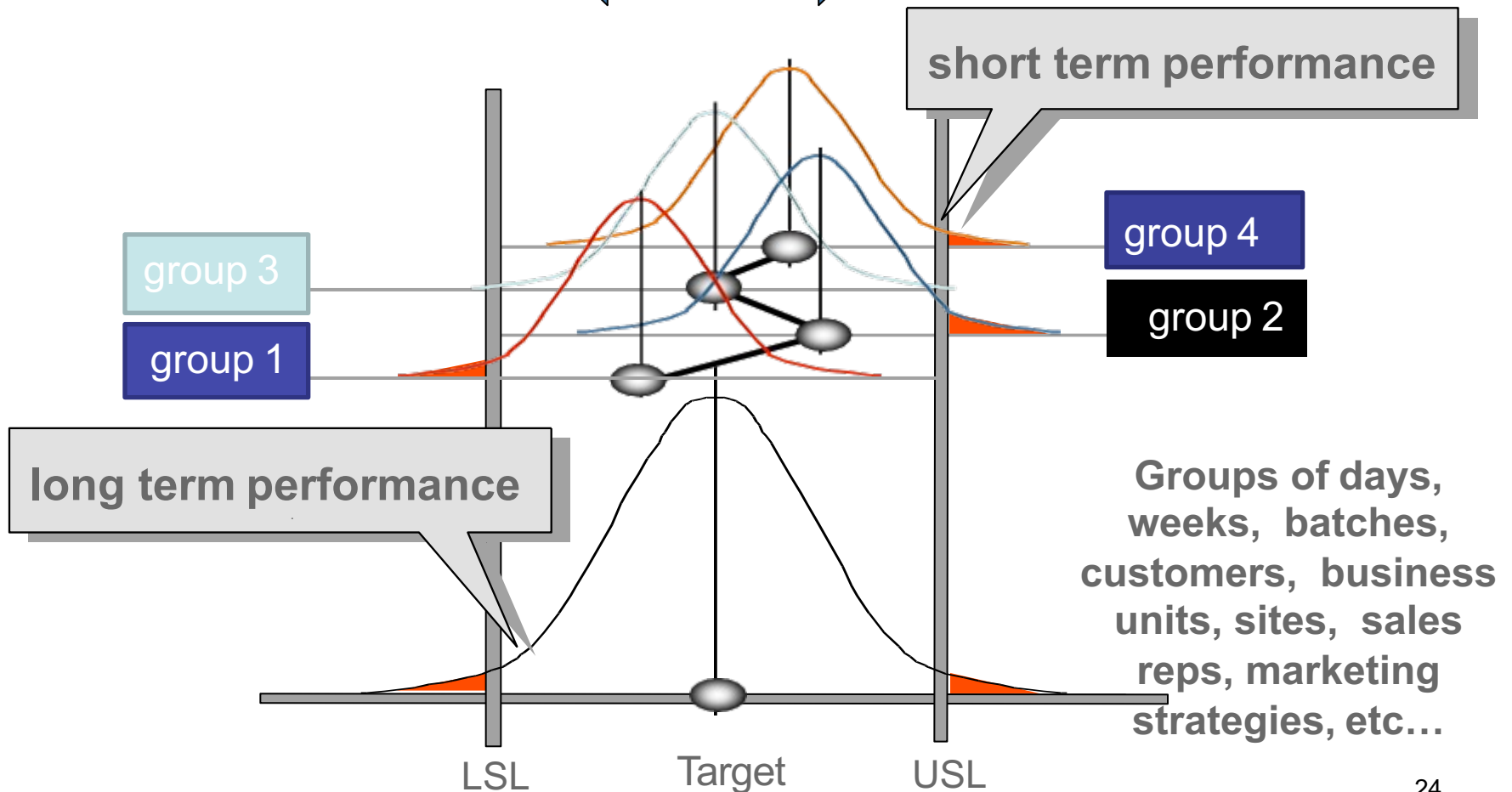


Teamwork will reduce process variation
and widen specification limits

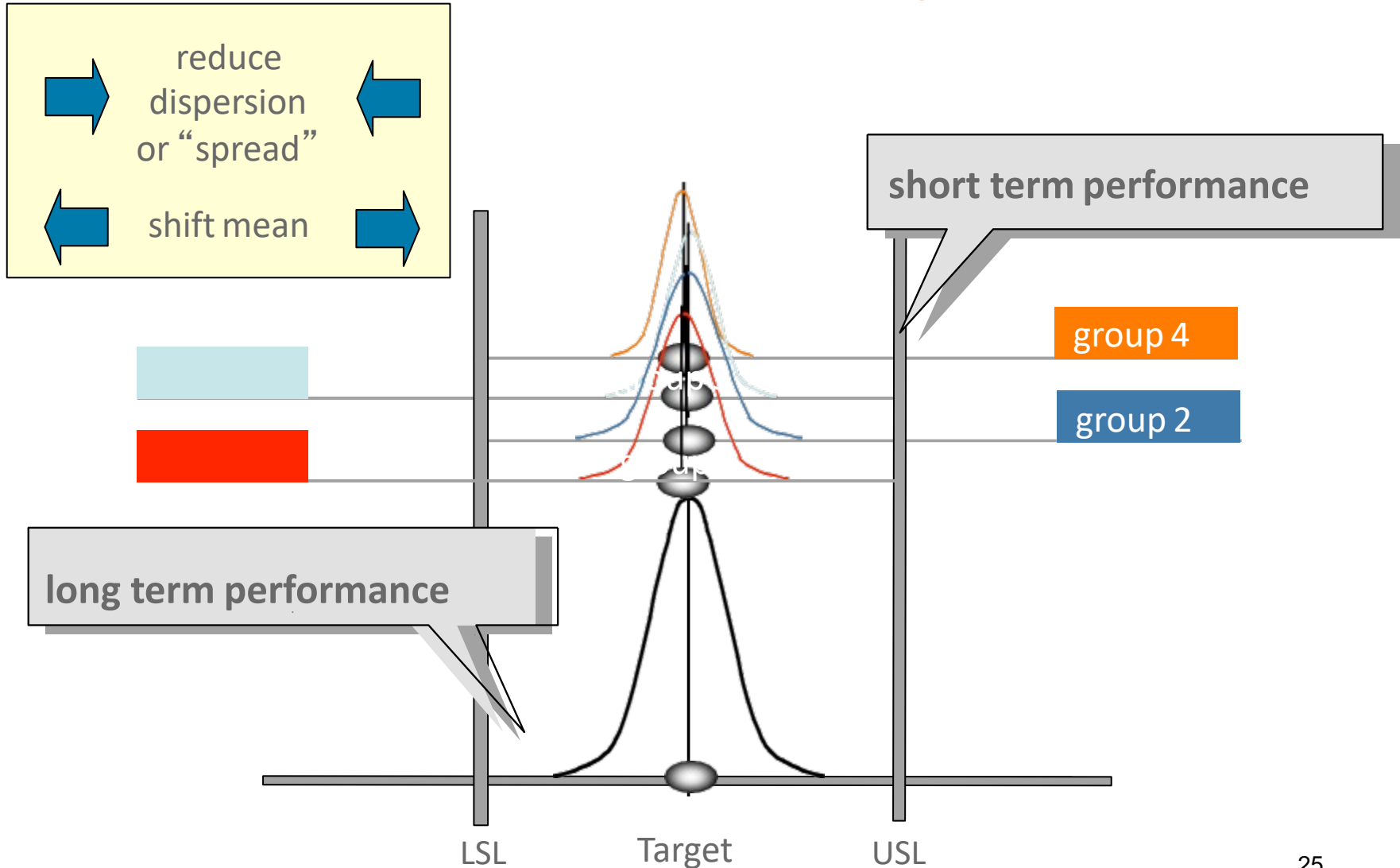
Basic statistic techniques and tools

- Do your own calculations, visualize data
- Clean your data (outliers, bad or missing data, ...)
- Excel – statistical formulas and functions
- Statistical packages:
 - R + R studio (free software)
 - Minitab
 - Matlab/Python
 - Statistica, SAS, ...
- A tool helps you in analysis phase of problem solving, awareness of statistics is important: you have to know what you are doing

Process Variation – Pre-Project



Process Variation – Post-Project



Process stability and capability must be improved in order to improve product and process quality

Basic laws of physics teach that:

- A stable process has a minimum dispersion
- Each process, running on its own, eventually becomes unstable
- The dispersion of such a process will increase in a natural but unavoidable way
- Eventually, there comes a time that we have problems due to process instability

Most processes can be described using statistical principles:

*If a characteristic of a product is the result of a process with only **natural** deteriorations, **then** the measurement distribution of this characteristic will follow a **normal distribution***

This forms the foundation of the **Statistical Process Control (SPC)**, which can be used to stabilize a process, thus improving the process capability.



A standard Normal Distribution is governed by:
$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Normal distribution

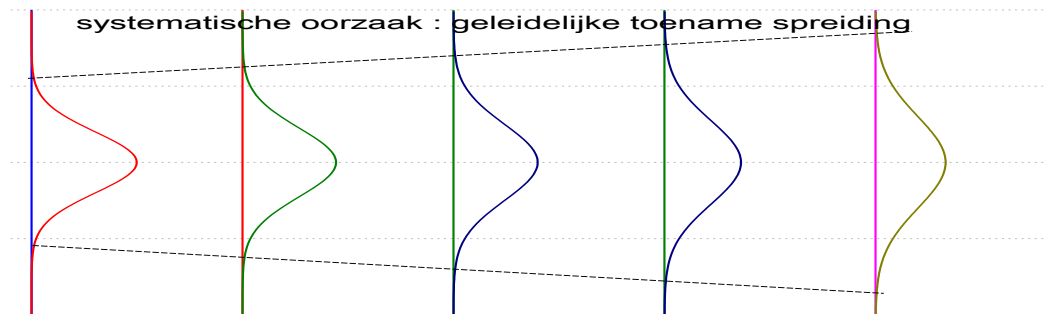
- Common distribution is normal distribution
Dispersion by natural/physical causes are normal distributed
- **Central limit theorem:**
 - Averages of different samples with sample number n are normal distributed
 - Spread = spread population / $\sqrt{\text{sample number } n}$
- There are also other distributions – examples
- **Visualize** data to see the distribution / outliers / ...
- Statistical calculation such as averages, spread ... only correct when the distribution is a normal distribution

Agenda

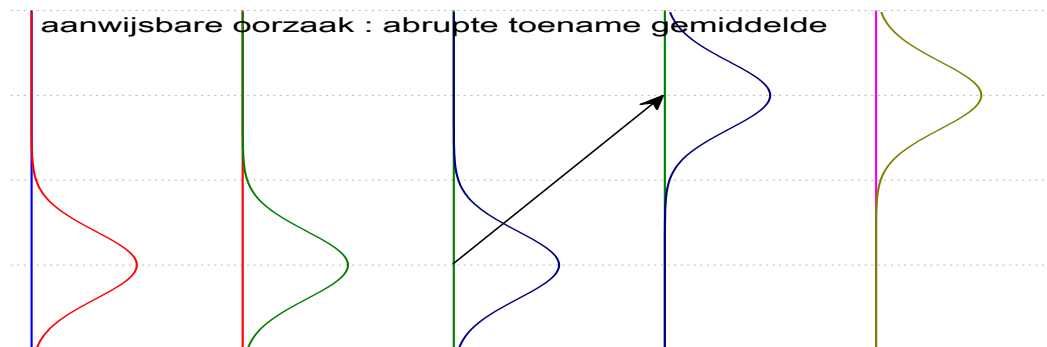
- 13u30-15u30: deel 1 Intro & **Process Capability**
- 15u30-15u45: pauze
- 15u45-17u45: deel 2 Statistical Process Control (charts)
- 17u45-18u15: pauze (broodjes)
- 18u15-20u15: deel 3 Exercises

Oorzaken van niet-stabiliteit

- Stabiliteit: proces enkel op toevallige wijze beïnvloed - proceskenmerken vertonen toevallige variaties
“common causes” versus “special causes”
- Systematische oorzaak : geleidelijk verloop distributie (ST)



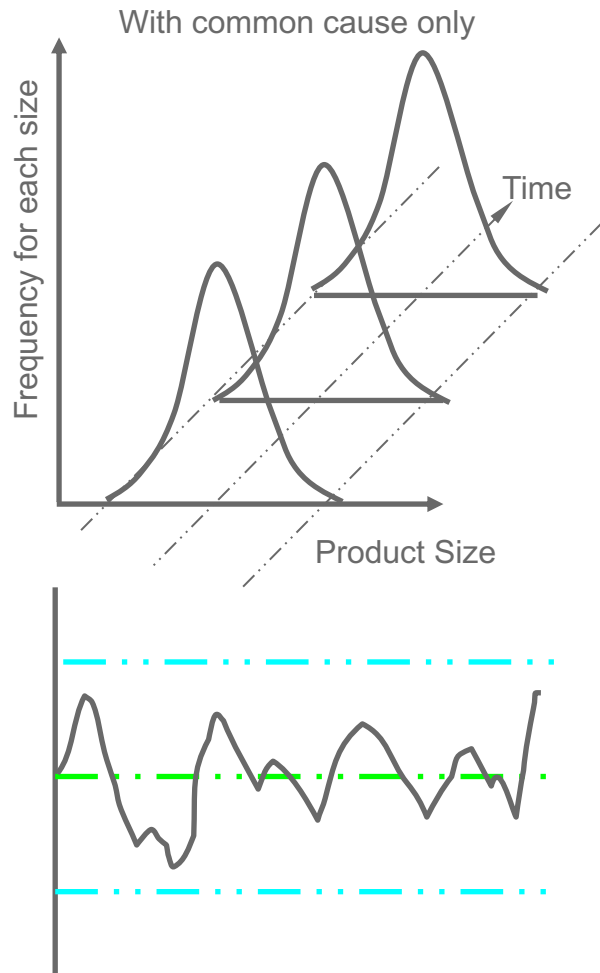
- Aanwijsbare oorzaak : abrupte wijziging distributie (LT)



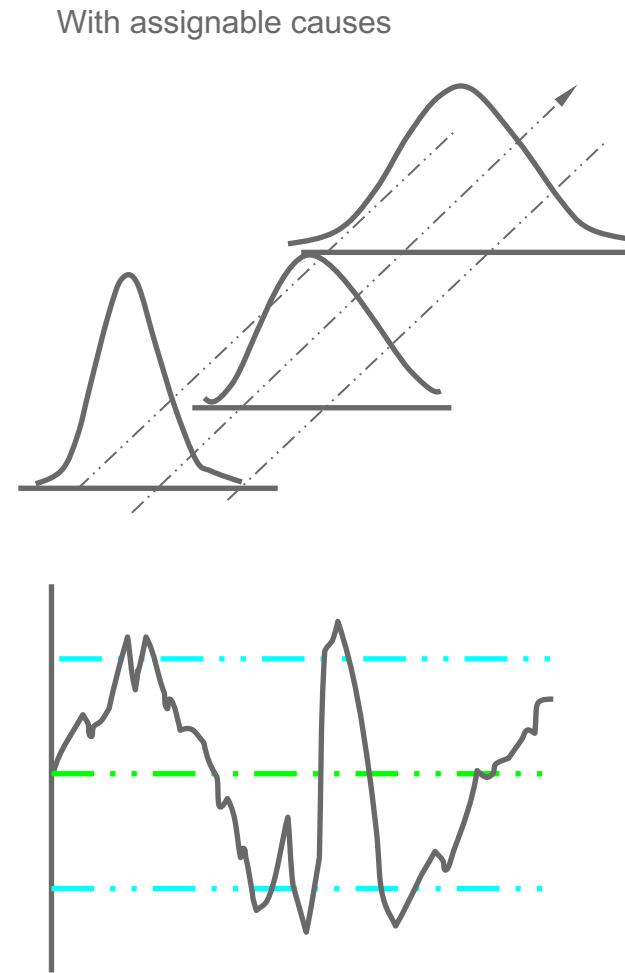
Common vs. Special Causes of Variation

Type of Variation	Definition	Typical Characteristics
Common Cause	Process is running within its normal operating range (Short term)	• Expected • Predictable range of values • Purely Random
Special Cause	Process displays variation outside what is expected (Long term)	• Unexpected • Unpredictable range of values • Not Purely Random

A process is said to be stable, predictable and in control if only common cause variability is present



- Consistent & stable
- Distribution can be predicted



- High variability
- Impossible to predict future condition

Process capability - C_p – C_{pk}

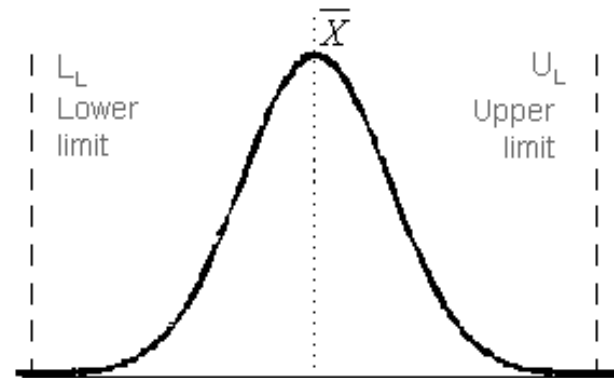
C = Capability, p = process, pk = process + Japans woord voor off-center

2 sided specification limits, one side: only C_{pk} can be calculated
(example Zn layer thickness)

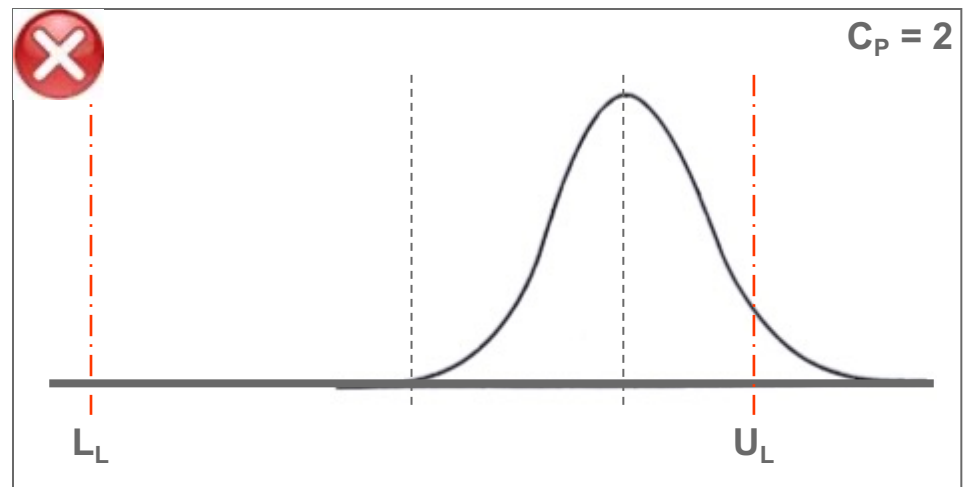
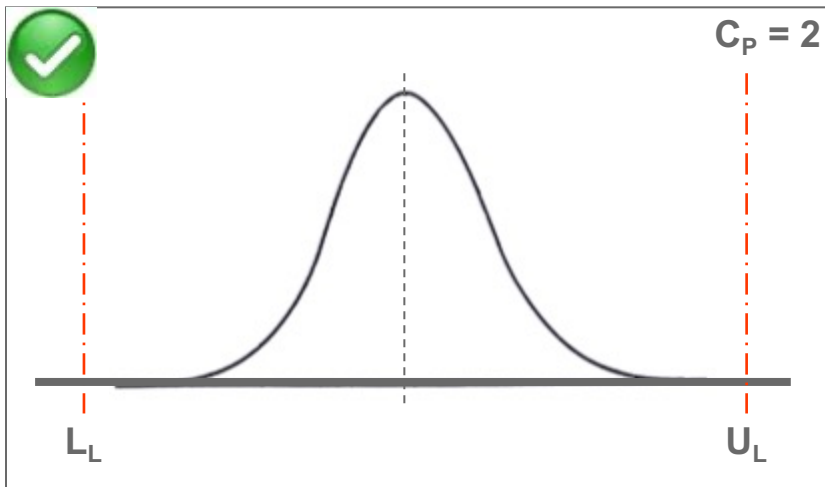
- C_p
Breedte van de distributie (spreiding) t.o.v. toegelaten interval,
Op zich niet voldoende.
(common cause – process engineer)
- C_{pk}
Ligging van gemiddelde t.o.v. toegelaten interval
(special cause – operator by doing adjustments)
- Korte termijn: C_p / C_{pk}
- Lange termijn (waarin de klant geïnteresseerd is): P_p / P_{pk}
P = Performance

The Process Capability (C_P) compares the process dispersion with the target given by the specification (U_L and L_L)¹

$$C_P = \frac{U_L - L_L}{6\sigma}$$



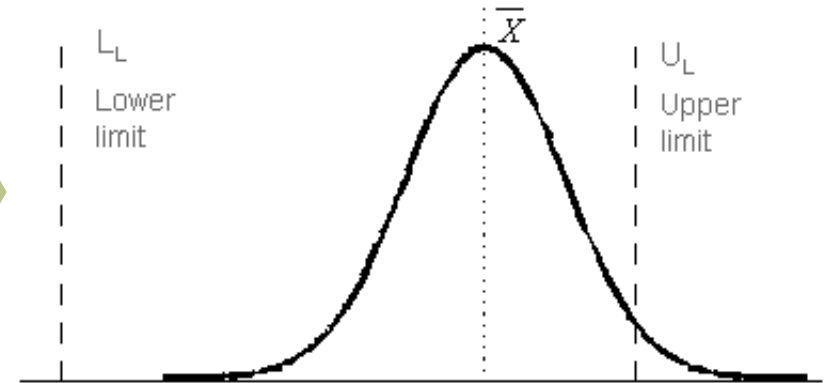
C_P is however not sufficient to define if a process is capable or not:



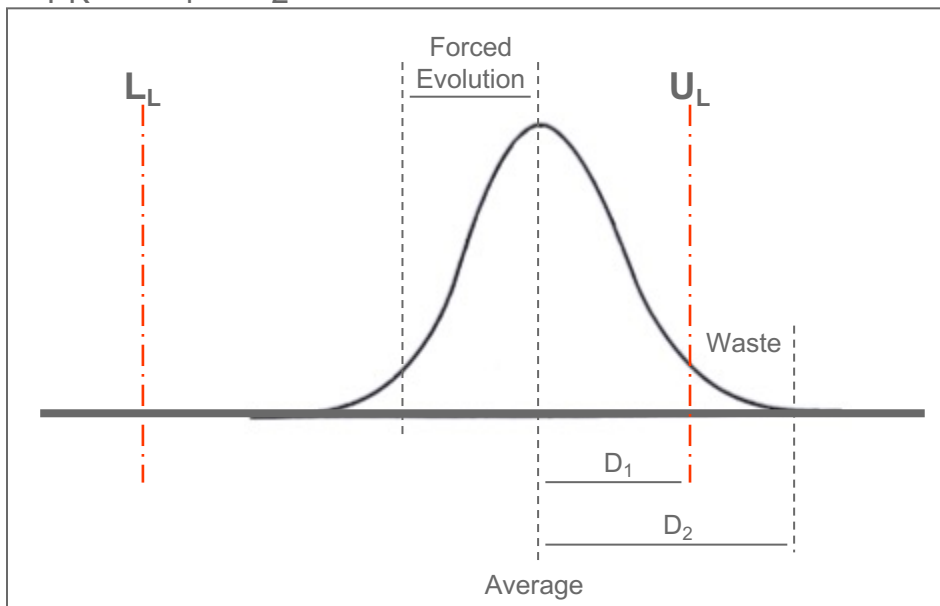
¹ U_L – Upper Limit; L_L – Lower Limit

The Process Capability (C_{PK}) compares the process dispersion with the target given by the specification

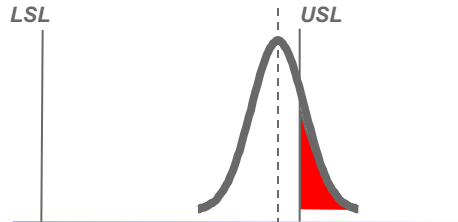
$$C_{pk} = MIN \left\{ \left(\frac{|U_L - \bar{X}|}{3\sigma} \right) \text{ and } \left(\frac{|\bar{X} - L_L|}{3\sigma} \right) \right\}$$



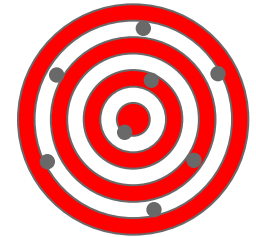
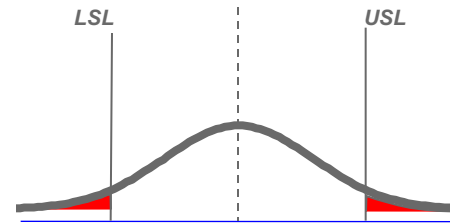
$$C_{PK} = D_1 / D_2$$



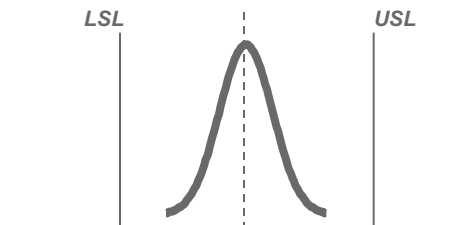
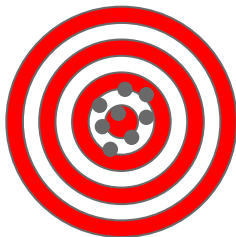
What do we Need?



Off-Target, Low Variation
High Potential Defects
Good C_p but Bad C_{pk}



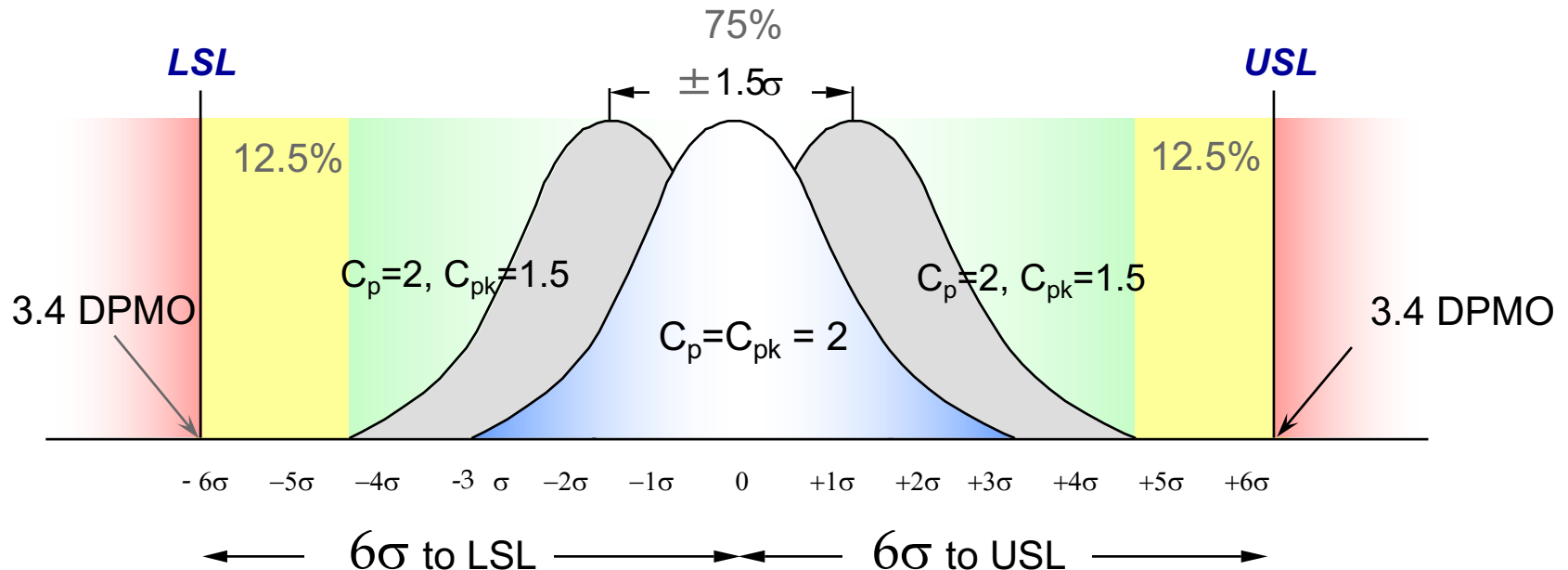
On Target
High Variation
High Potential Defects
No so good C_p and C_{pk}



On-Target, Low Variation
Low Potential Defects
Good C_p and C_{pk}

- Variation reduction and process centering create processes with less potential for defects.
- The concept of defect reduction applies to **ALL** processes (not just manufacturing)

Technical Definition of Six Sigma

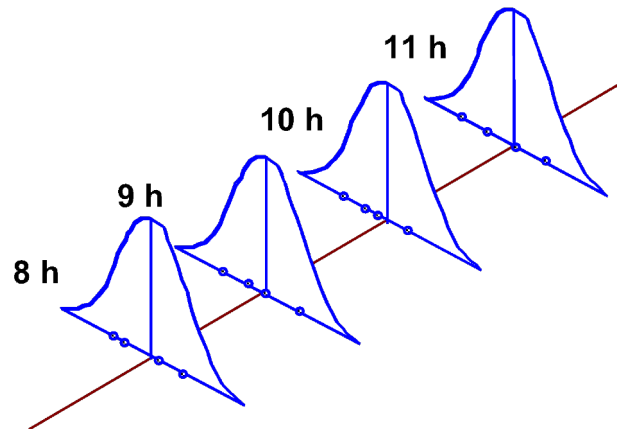


Evolution to:

A Management driven, scientific methodology for product and process improvement which creates breakthroughs in financial performance and Customer satisfaction

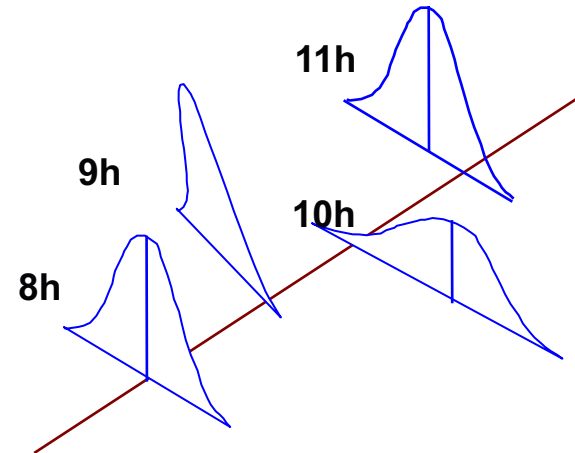
Process stability influences performance, predictability and capability

Stable & Predictable



Capable

Unstable & Unpredictable



Not capable

Some notes on Process stability

- An unstable process is unpredictable; a stable process is predictable (not on a piece by piece level, but over time, in the sense like described in the previous slide).
- Stability (and hence predictability) comes from common causes of variation; lack of stability (and hence no predictability) comes from special causes of variation.
- The ideal situation is a stable and sufficiently capable process.
So both special and common causes under control !



The Process Capability (C_P & C_{PK}) can be used to predict the capability of a process to produce a certain product

If the deviation is centred between the lower and upper limits ($C_P = C_{PK}$):

$$C_P < 1$$

The process is not capable to produce products properly.

$$C_P = 1$$

The process is just capable to produce products properly. About 2700 items out of 10^6 items are out of tolerances.

$$\text{Specification} = U_L - L_L = 6\sigma \rightarrow 99.73\%$$

$$C_P \geq 1.33$$

The capability of the process is acceptable. About 60 items out of 10^6 items are out of tolerances (64ppm).

$$\text{Specification} = U_L - L_L = 8\sigma \rightarrow 99.9936\%$$

$$C_P \geq 1.67$$

The capability of the process is good. About 0,57 items out of 10^6 items are out of tolerances.

$$\text{Specification} = U_L - L_L = 10\sigma \rightarrow 99.999943\%$$

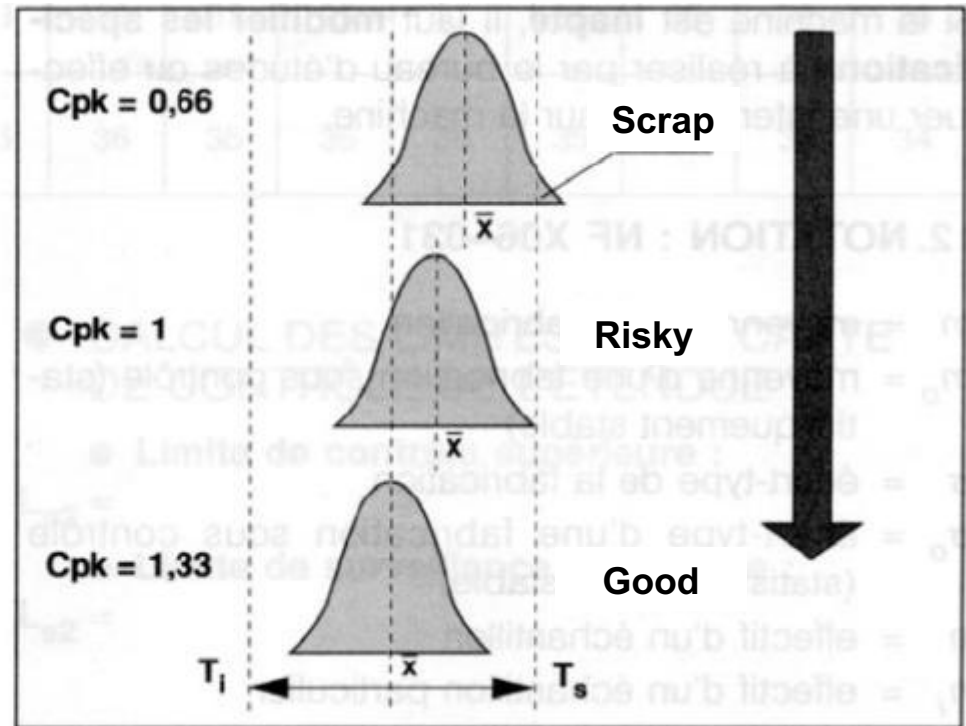
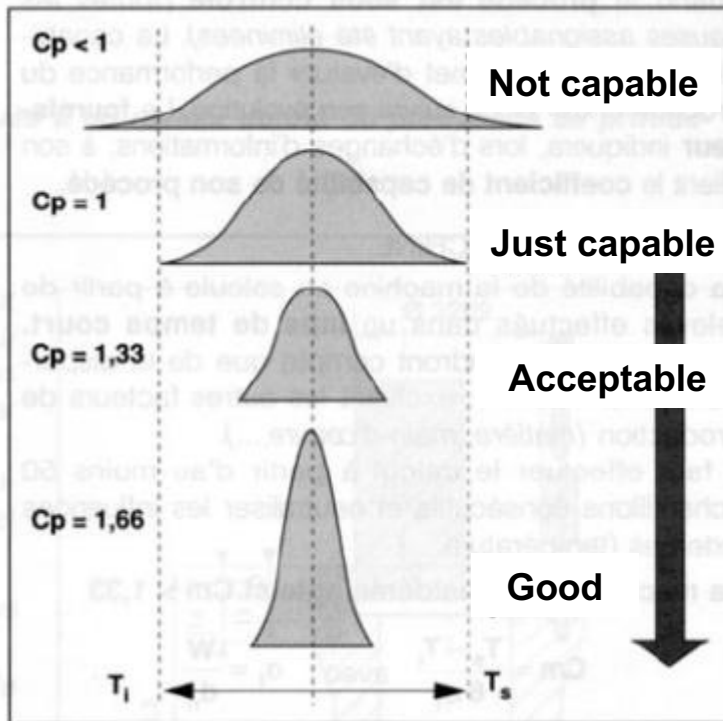
$$C_P = 2$$

6 Sigma quality level – 2 defect per billion of opportunities (short term)

The Process Capability (C_P & C_{PK}) can be used to predict the capability of a process to produce a certain product

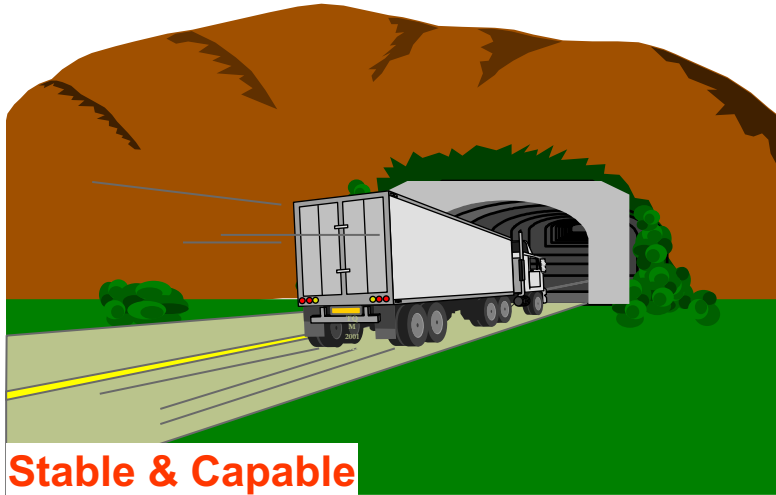


ArcelorMittal

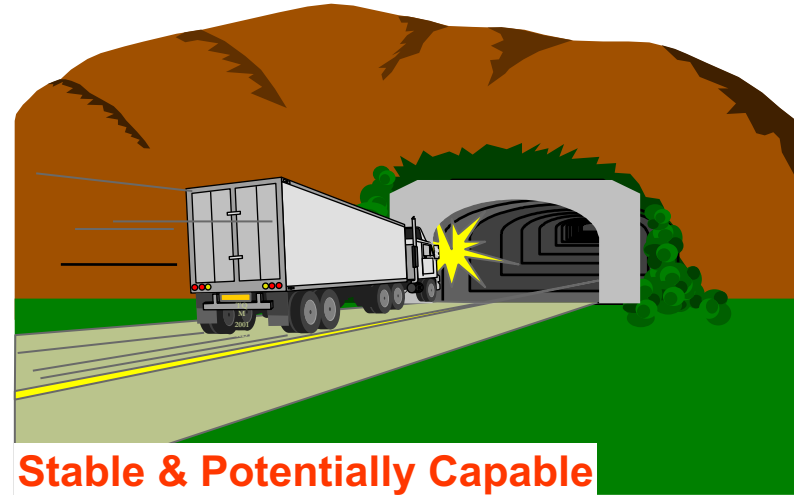


The Process Capability (C_P & C_{PK}) can be used to predict the capability of a process to produce a certain product

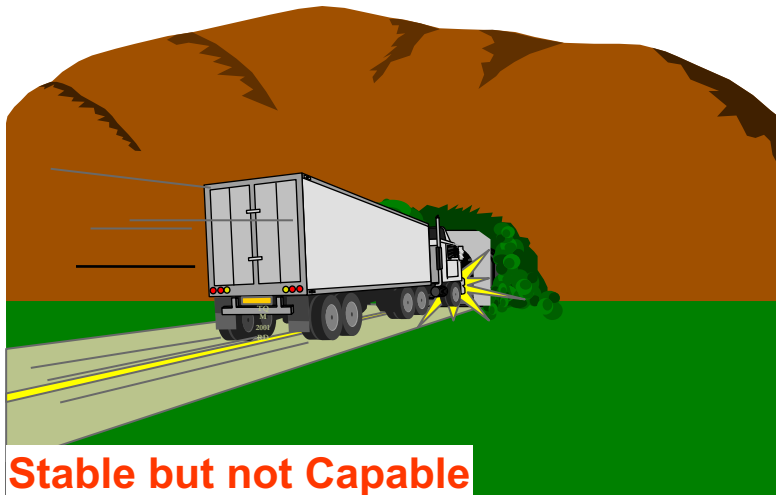
Good C_P and Good C_{PK}



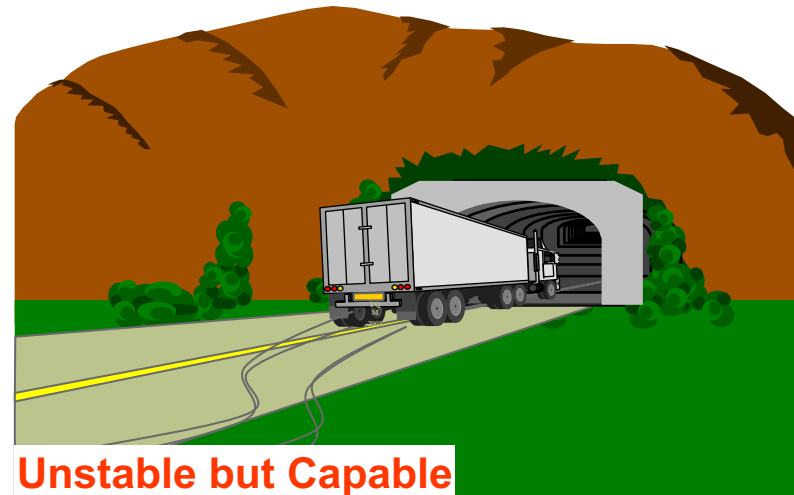
Good C_P but bad C_{PK} – Adjustment needed



Bad C_P and bad C_{PK} – Long term problems



Good C_P and good C_{PK} – Acceptable (for now...)

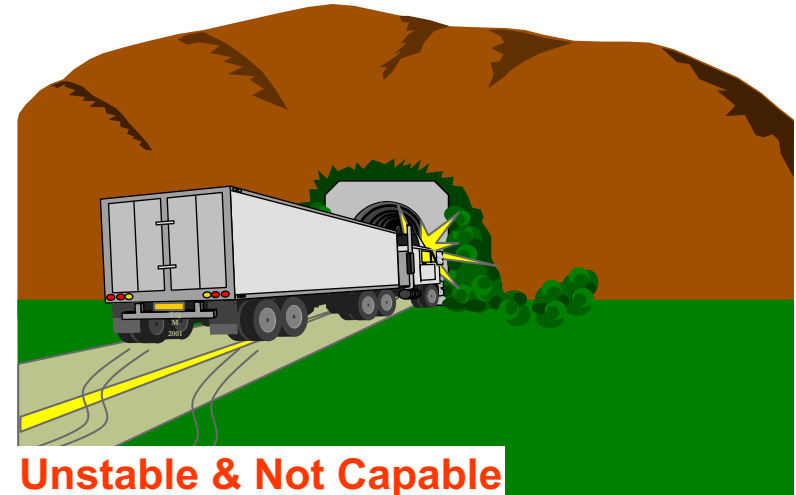


The Process Capability (C_P & C_{PK}) can be used to predict the capability of a process to produce a certain product

Good C_P and bad C_{PK} – Urgent action needed



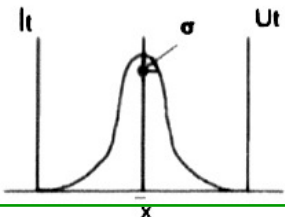
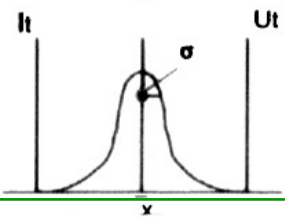
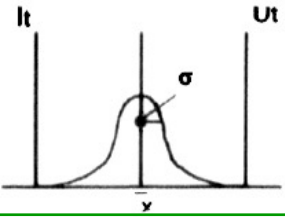
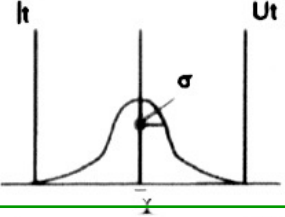
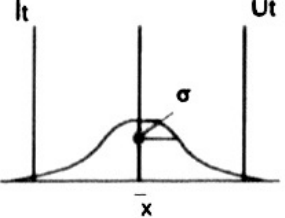
Bad C_P and bad C_{PK} – not desperate but...



The table below can be used as a guide when considering process capability

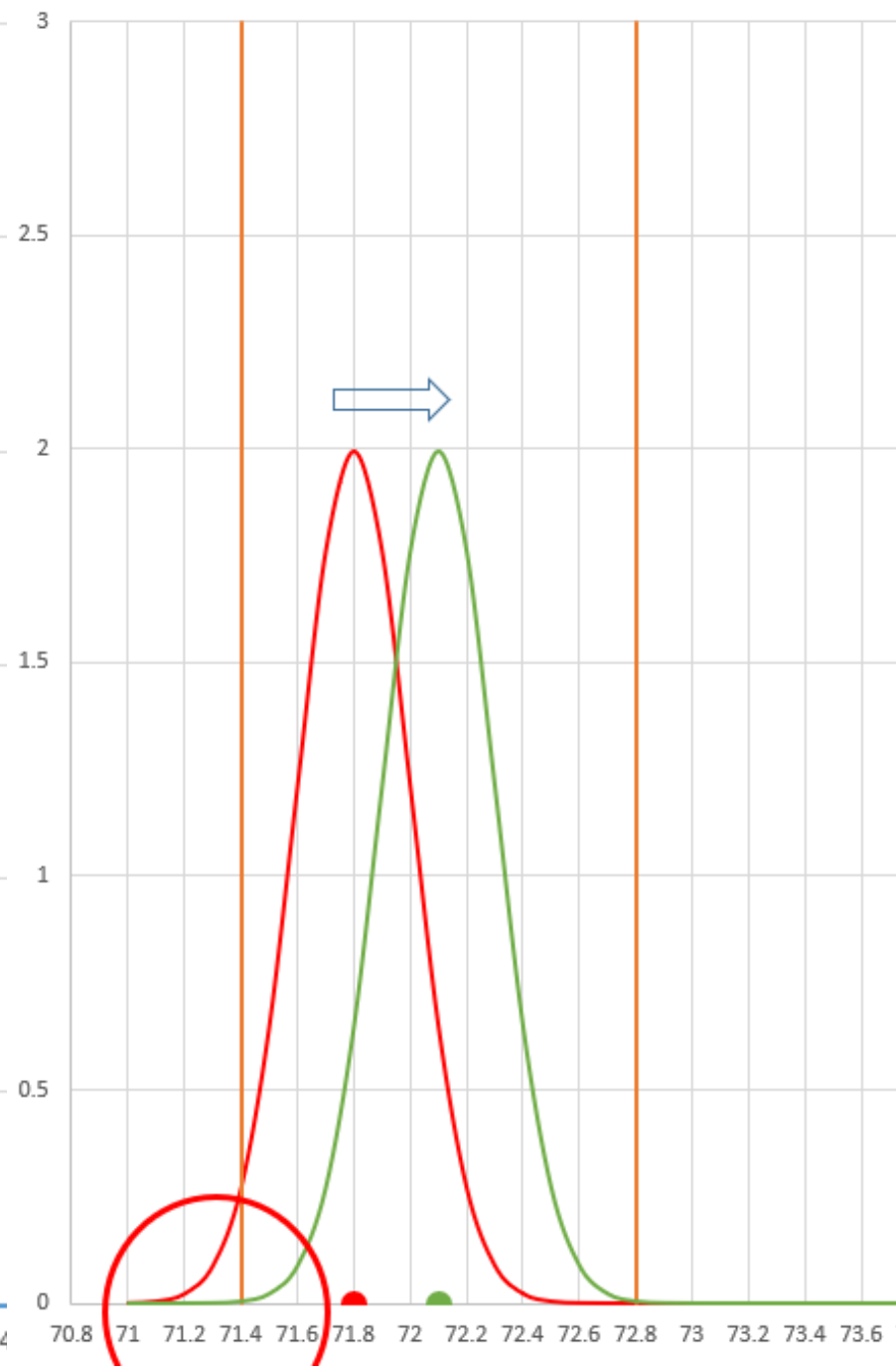
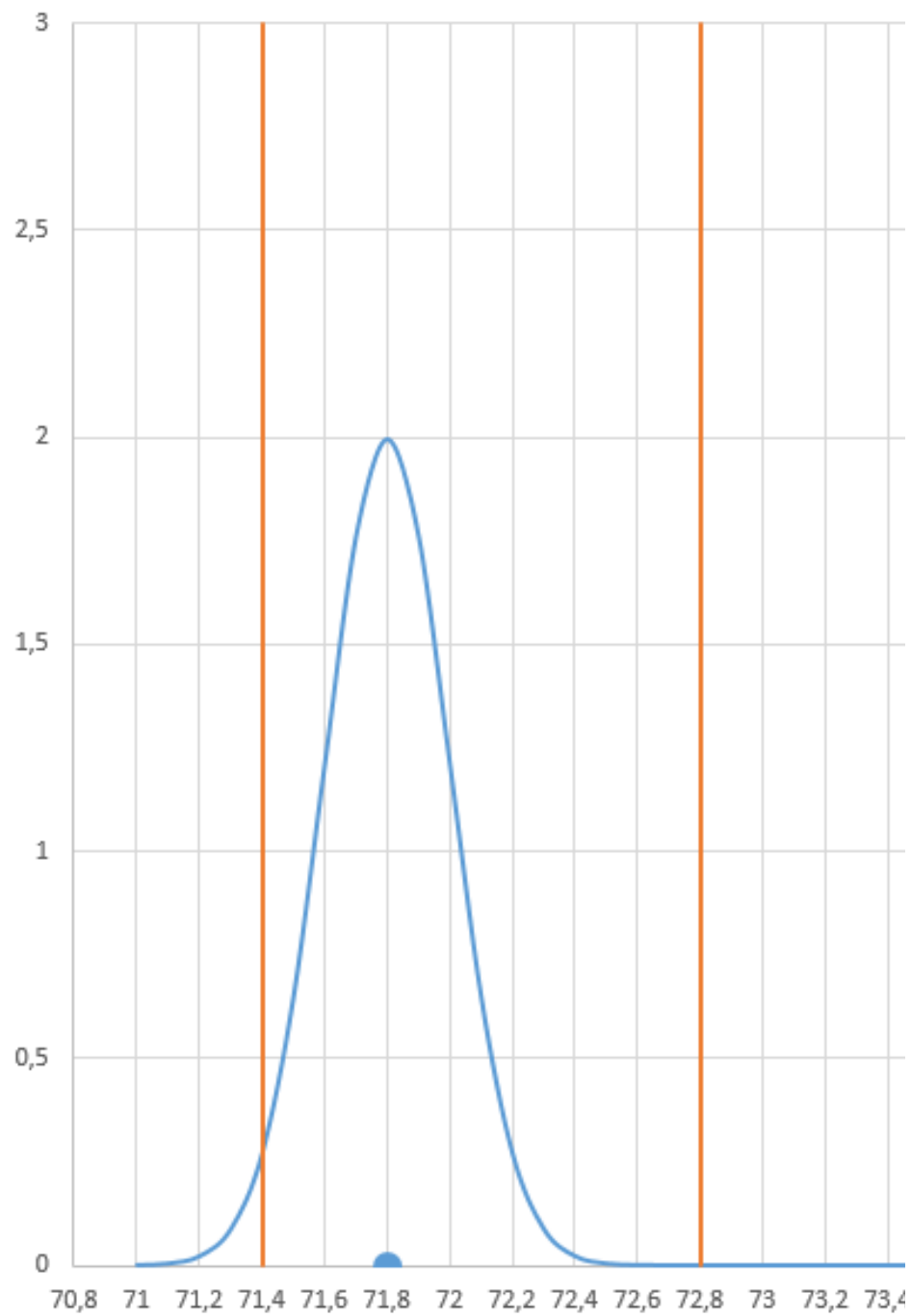
	Parameter			Comments
	Stability	Capability C_P	Capability C_{PK}	
Process Status	+	+	+	Keep it this way
	+	+	-	Simple question of adjustment (operator)
	+	-	+	Impossible
	+	-	-	Long term and difficult solutions (engineer)
	-	+	+	The first blunder...
	-	+	-	Urgent reaction needed...
	-	-	+	Impossible
	-	-	-	Not desperate but ...

The table below can be used as a guide when considering process capability

C_p	Distribution	Judgment	Action
$C_p \geq 1.67$		Process capability is more than enough.	- Simplification of process control and cost reduction can be considered in certain cases. ✓ Quarterly sampling on products
$1.67 > C_p \geq 1.33$		Process capability is sufficiently high.	- Ideal condition. - Maintain it. ✓ Regular sampling on products
$1.33 > C_p \geq 1.00$		Process capability is not sufficiently high, but is adequate.	- Control process properly and maintain it in a control state. - Defects may result if C_p approaches 1 - Take action if needed. ✓ Frequent sampling on products
$1.00 > C_p \geq 0.67$		Process capability is not sufficient.	- Defects have been generated. - Screening inspection, process control and Kaizen will be required. ✓ All products must be checked
$0.67 > C_p$		Process capability is very low.	- Cannot satisfy quality. Quality must be improved, cause must be pursued and emergency actions must be taken. - Reexamine standards. ✓ All product must be checked

Proces capabiliteit: oefening

- Een as die wordt gebruikt in een automatische transmissie moet een lengte hebben tussen 71,4 mm en 72,8 mm. Het productieproces wordt gemonitord met een Xbar-kaart en een s-kaart, met steekproeven van lengte 5
- De kaarten tonen aan dat het proces beheerst verloopt. Uit de kaarten blijkt dat $\bar{X} = 71,8$ mm en $\bar{s} = 0,2$ mm
- Is dit proces geschikt om aan de specificaties van de klant te voldoen?
- Hoeveel % uitval mogen we verwachten?
- Wat als gemiddelde in het midden gelegd wordt? Wat wordt dan C_p ?, C_{pk} ?, uitval?



Minitab capability example

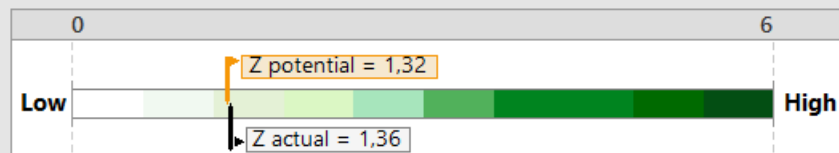
Capability Analysis for C1 Report Card

Check	Status	Description
Stability		Stability is an important assumption of capability analysis. Although no points are outside the control limits, there are too many points in a row within 1 standard deviation of the center line on the Xbar chart. The within-subgroup standard deviation is most likely inflated, which affects the Potential capability estimates. In addition, the limits on the Xbar chart are too wide, making it difficult to assess stability. This condition is likely caused by a systematic source of variation within the subgroups. Examine your data collection strategy to determine possible sources.
Number of Subgroups		You only have 13 subgroups. For a capability analysis, it is generally recommended that you collect at least 25 subgroups over a long enough period of time to capture the different sources of process variation.
Normality		Your data passed the normality test. As long as you have enough data, the capability estimates should be reasonably accurate.
Amount of Data		The total number of observations is 100 or more. The capability estimates should be reasonably precise.

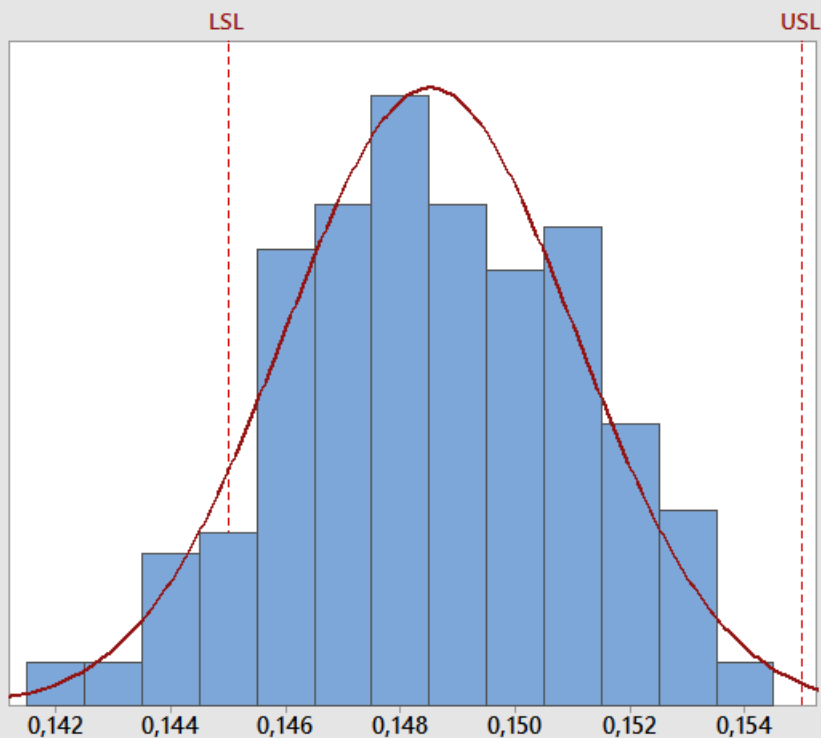
Capability Analysis for C1

Summary Report

How capable is the process?



Actual (Overall) Capability
Are the data inside the limits?



Customer Requirements

Upper spec	0,155
Target	*
Lower spec	0,145

Process Characterization

Mean	0,14852
Standard deviation (overall)	0,0025303
Actual (overall) capability	
Pp	0,66
Ppk	0,46
Z.Bench	1,36
% Out of spec	8,74
PPM (DPMO)	87445

Comments

- The defect rate is 8,74%, which estimates the percentage of parts from the process that are outside the spec limits.

Actual (overall) capability is what the customer experiences.

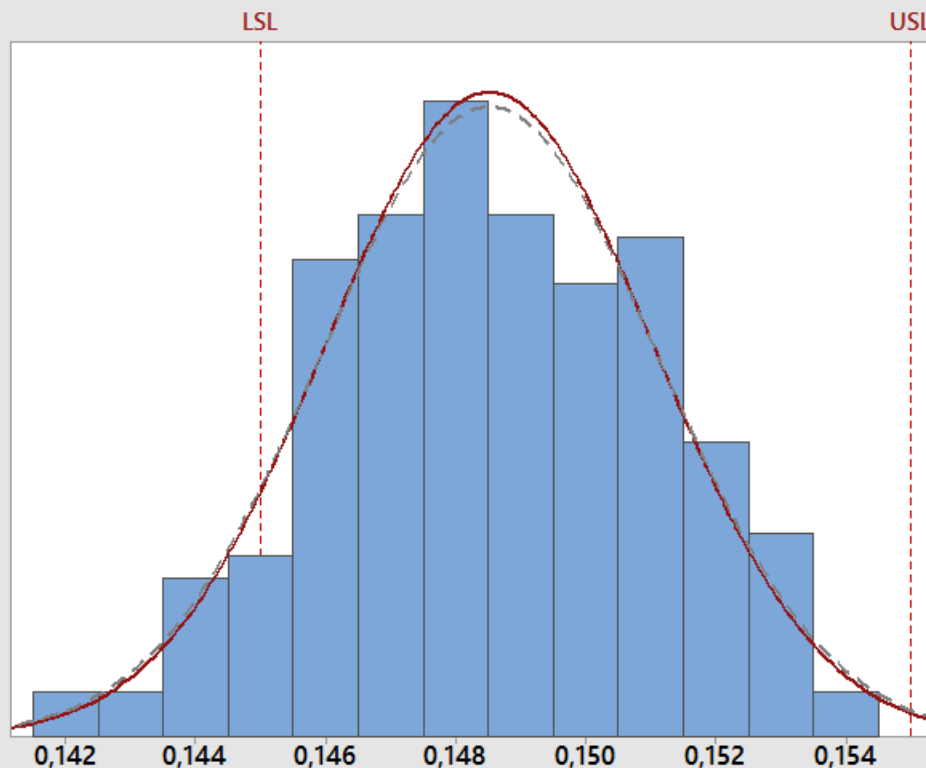
Potential (within) capability is what could be achieved if process shifts and drifts were eliminated.

Capability Analysis for C1

Process Performance Report

Capability Histogram

Are the data inside the limits?



— Actual (overall) capability is what the customer experiences.

- - - Potential (within) capability is what could be achieved if process shifts and drifts were eliminated.

Process Characterization

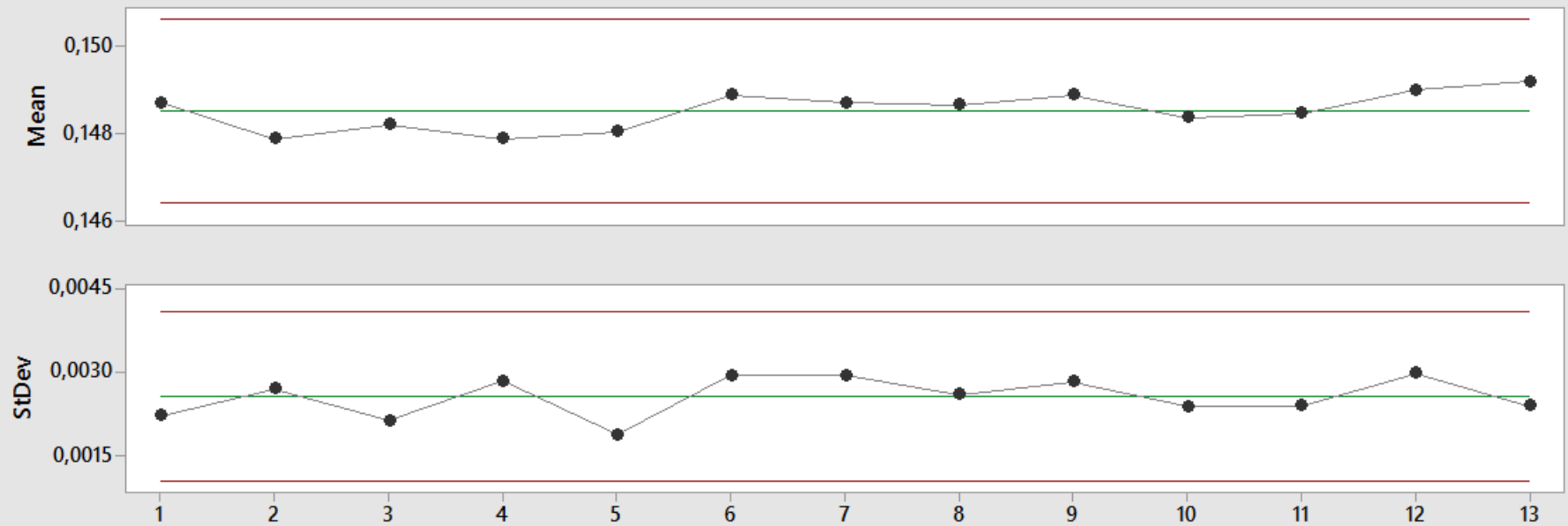
Total N	180
Subgroup size	14
Mean	0,14852
Standard deviation (overall)	0,0025303
Standard deviation (within)	0,0025891

Capability Statistics

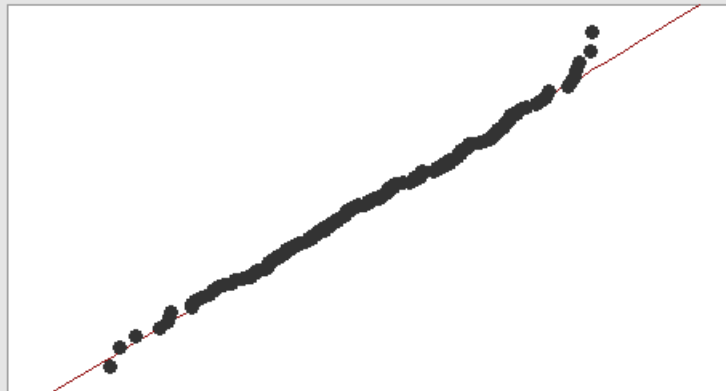
Actual (overall)	
Pp	0,66
Ppk	0,46
Z.Bench	1,36
% Out of spec (observed)	8,89
% Out of spec (expected)	8,74
PPM (DPMO) (observed)	88889
PPM (DPMO) (expected)	87445
Potential (within)	
Cp	0,64
Cpk	0,45
Z.Bench	1,32
% Out of spec (expected)	9,33
PPM (DPMO) (expected)	93282

Capability Analysis for C1 Diagnostic Report

Xbar-S Chart
Confirm that the process is stable.



Normality Plot
The points should be close to the line.



Normality Test
(Anderson-Darling)

Results	Pass
P-value	0,553

Agenda

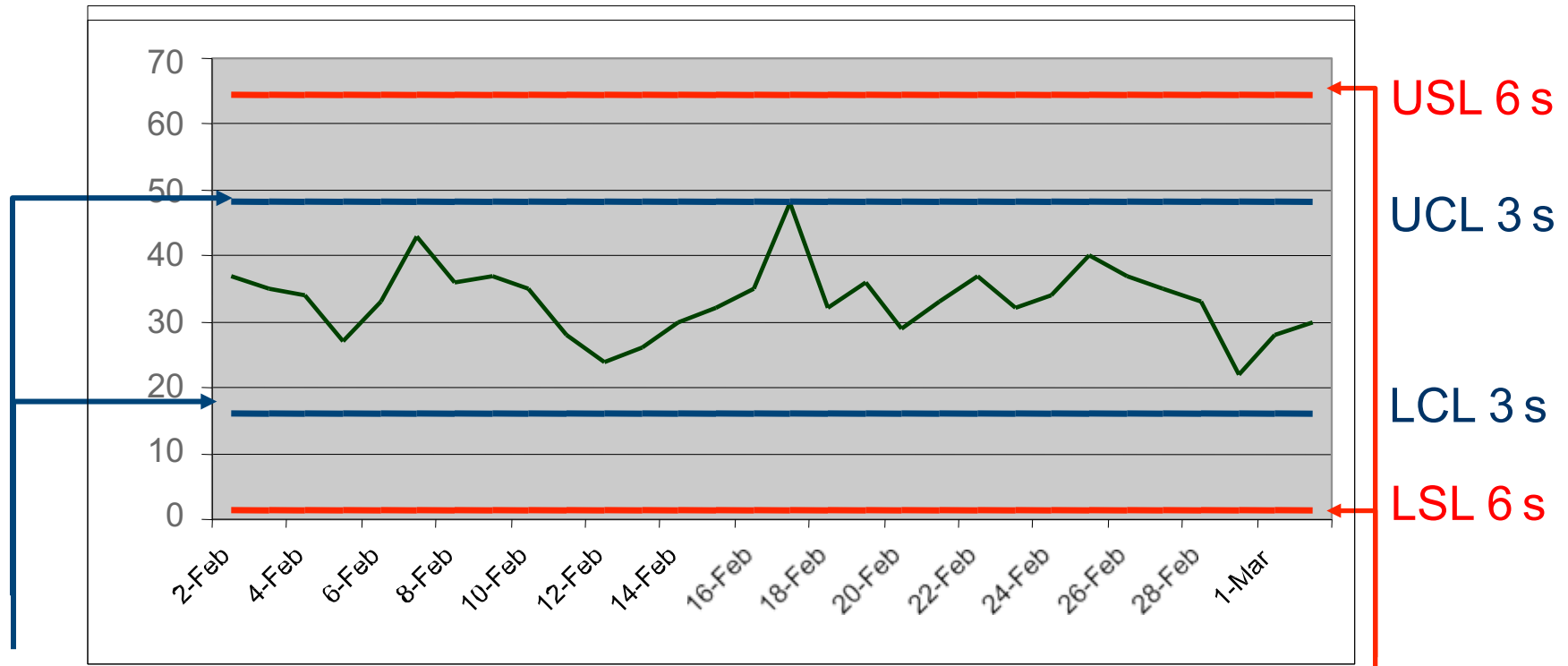
- 13u30-15u30: deel 1 Intro & Process Capability
- 15u30-15u45: pauze
- 15u45-17u45: deel 2 Statistical Process Control (charts)
- 17u45-18u15: pauze (broodjes)
- 18u15-20u15: deel 3 Exercises

SPC – Control charts

Wat is SPC ?

- SPC = Statistical Process Control
(Statistische Procesbeheersing)
- Techniek om stabiliteit productieproces te bewaken
(bescherming tegen aanwijsbare, niet tegen systematische oorzaken)
- Stabiel proces = distributies proceskenmerken zijn constant over de tijd (bv. $N(\mu, \sigma)$ met μ en σ constant)
- Opvolging d.m.v. regelmatige controles van essentiële proceskenmerken (= statistische fase)
- Indien controle NOK dan actie (= engineering fase)

Control Limits are Not Specifications



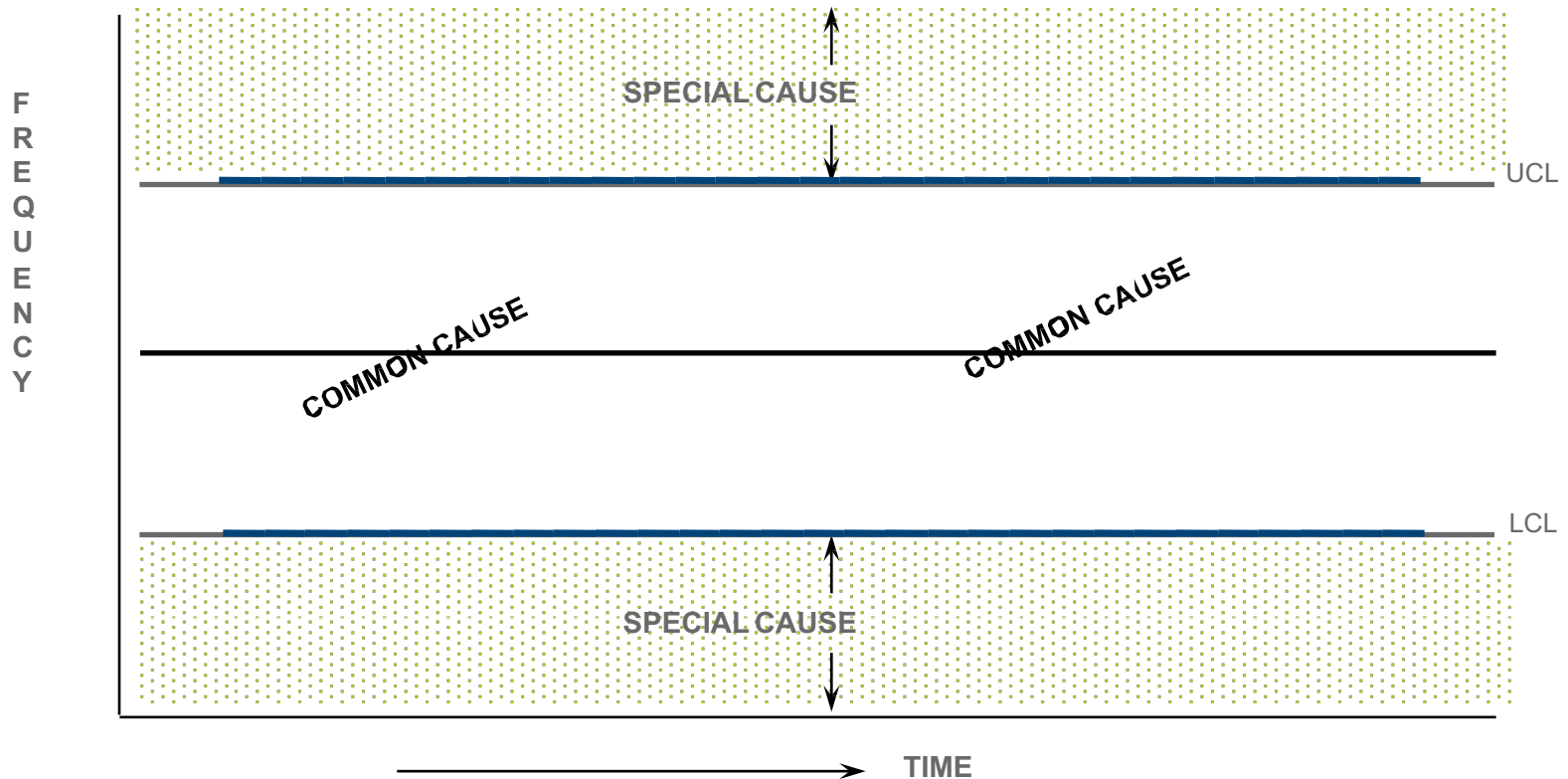
Control Limits (UCL, LCL)

- Defined based on process performance (± 3 estimated standard deviations from the mean)
- Help determine if your process is “in control” (without special cause variation)
- Plotted on control charts
- Represent the **voice of the process**

Customer Specification Limits (USL, LSL)

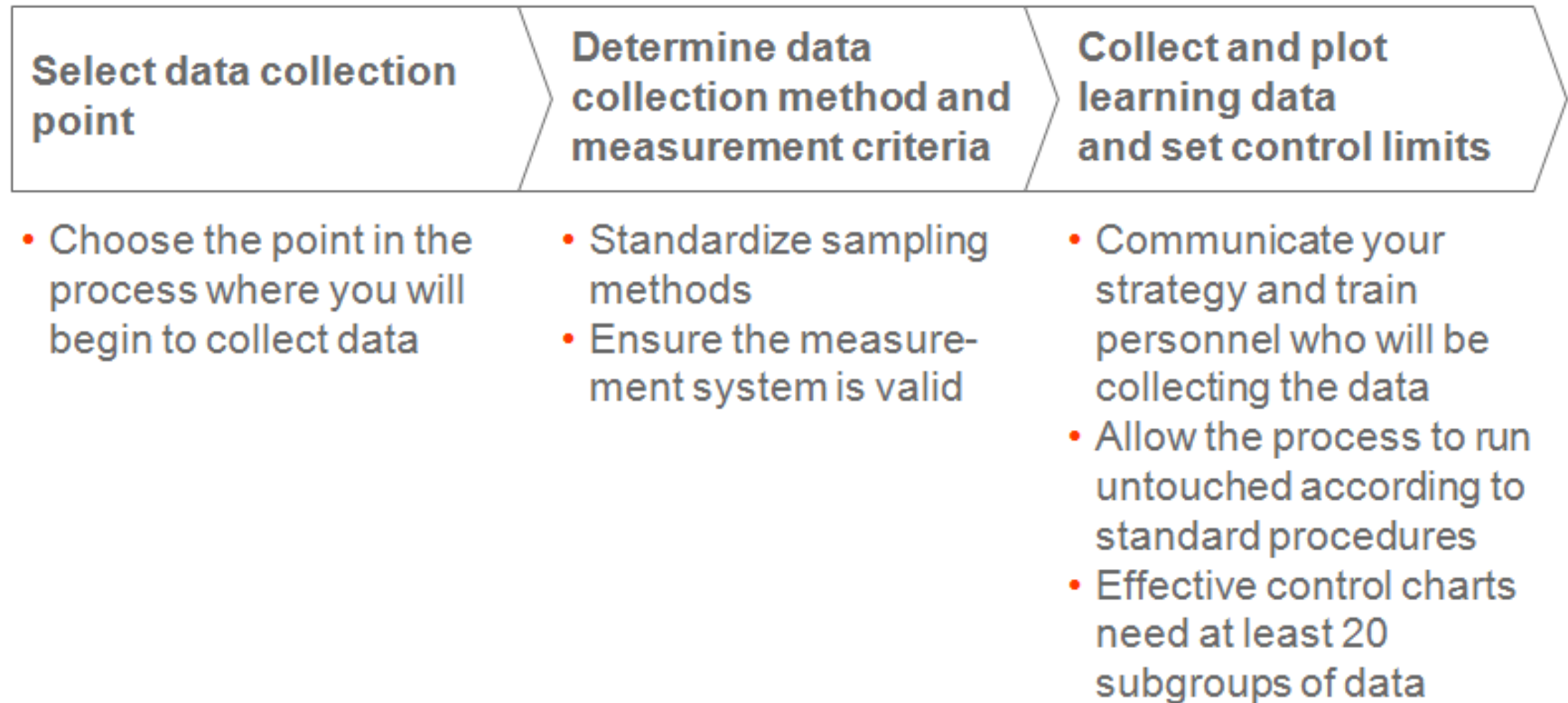
- Defined based on feedback from the customer(s)
- Help determine if your process is producing defects
- Plotted on histograms (not control charts)
- Represent the **voice of the customer**

Purpose for Using Control Charts

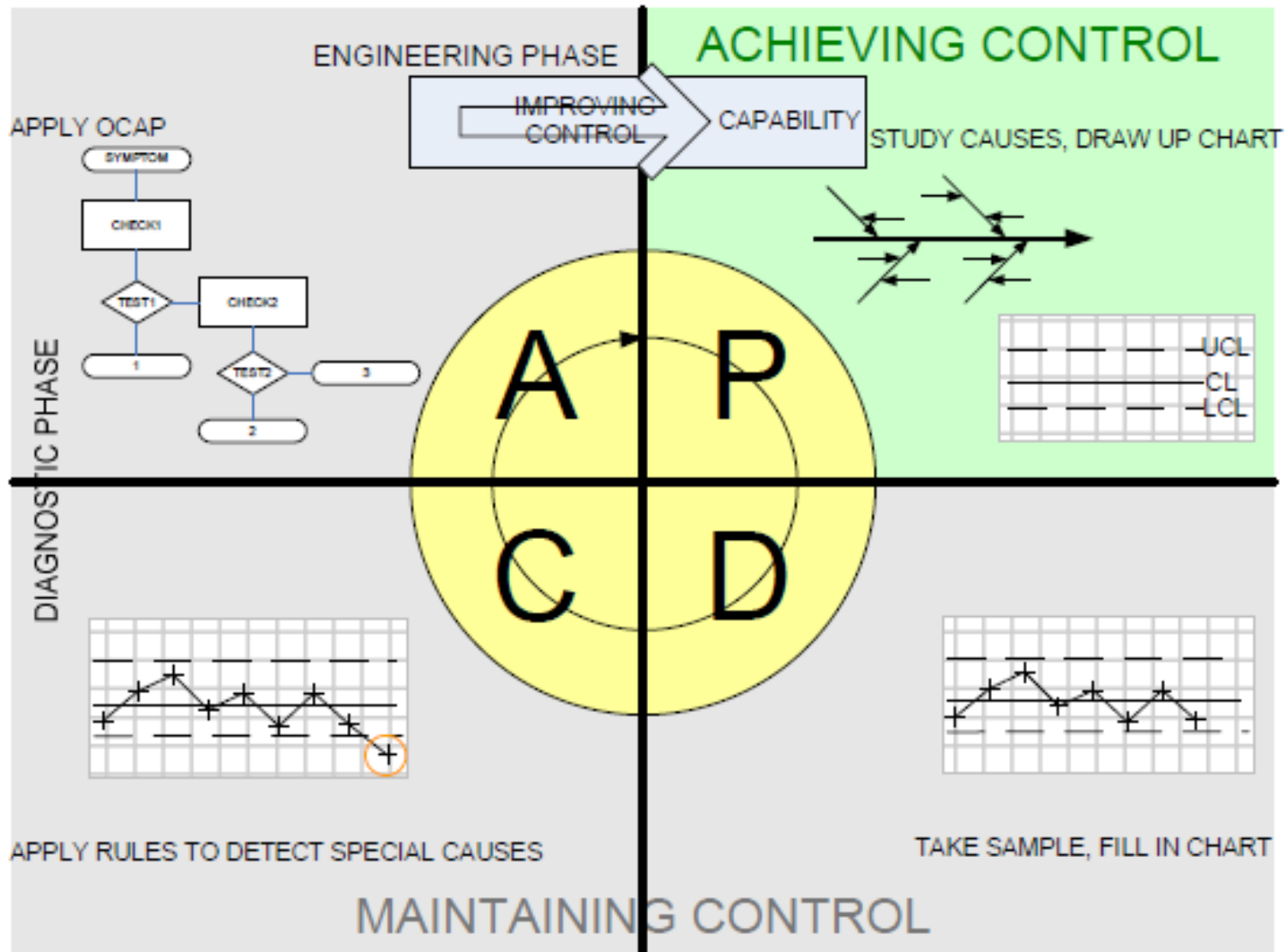


The Control Limits probabilistically separate common cause and special cause variability.

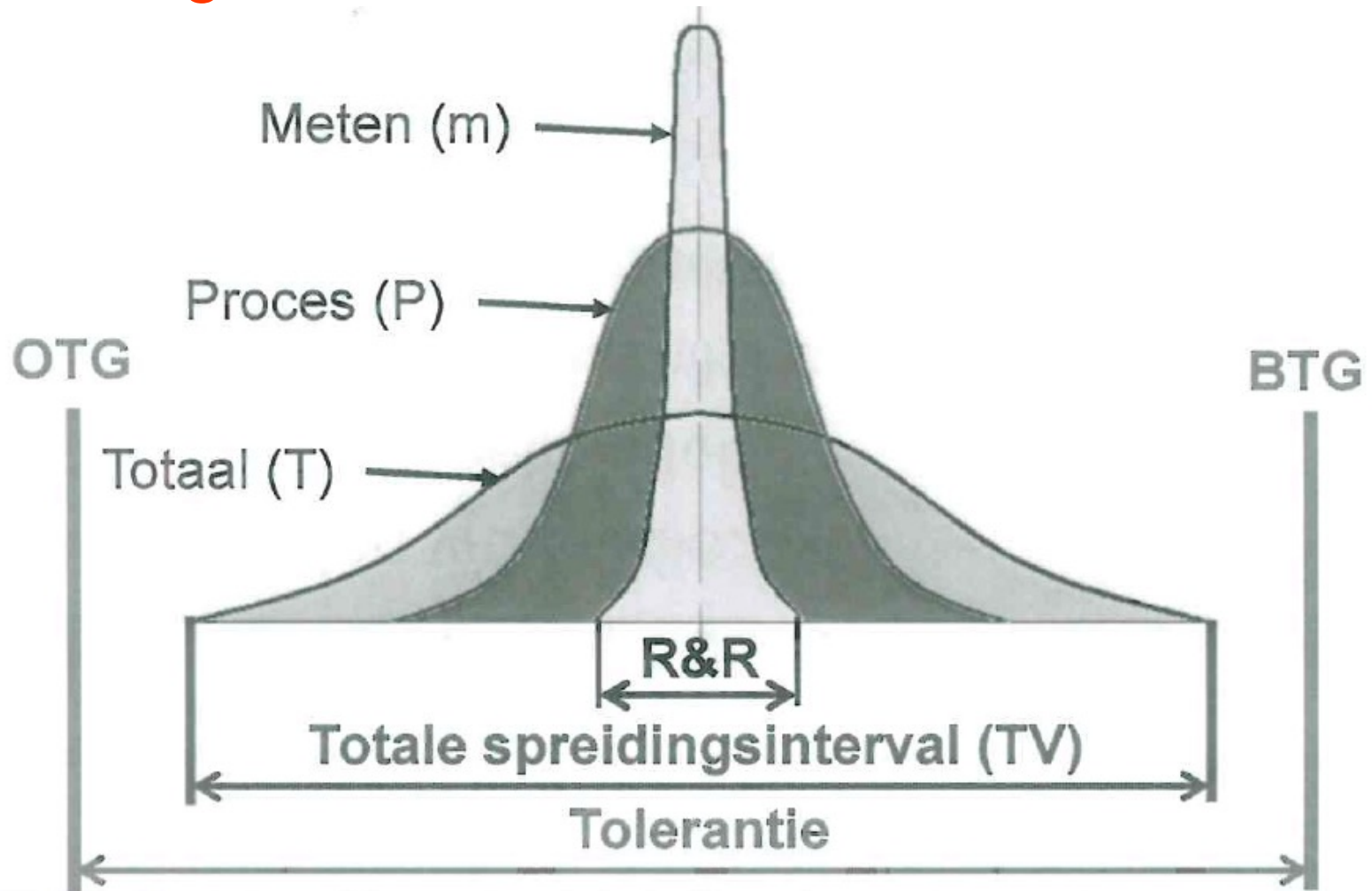
SPC process set up



SPC: de PDCA-cirkel



Spreading door meetinstrument

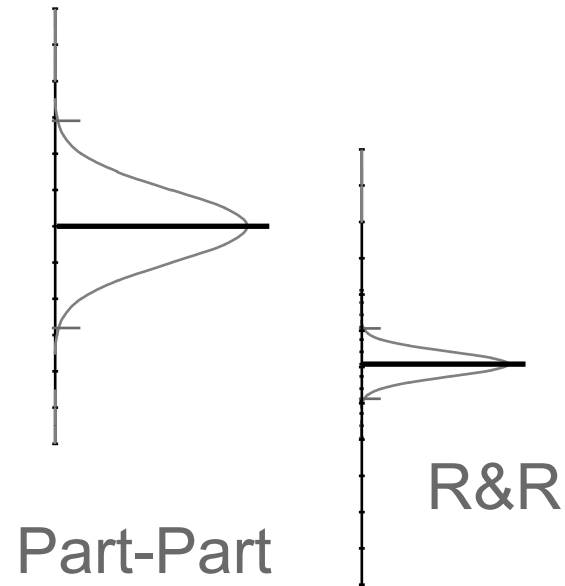


Gauge R&R (MSA) impact measurement gauge

$$\sigma^2_{\text{Total}} = \sigma^2_{\text{Part-Part}} + \sigma^2_{\text{R\&R}}$$

Recommendation:

Resolution $\leq 10\%$ of tolerance to measure
Gauge R&R $\leq 20\%$ of tolerance to measure



- **Repeatability (Equipment variation)**

Variation observed with one measurement device when used several times by **one operator** while measuring the identical characteristic on the same part.

- **Reproducibility (Appraised variation)**

Variation obtained from **different operators** using the same device when measuring the identical characteristic on the same part.

- **Stability or Drift**

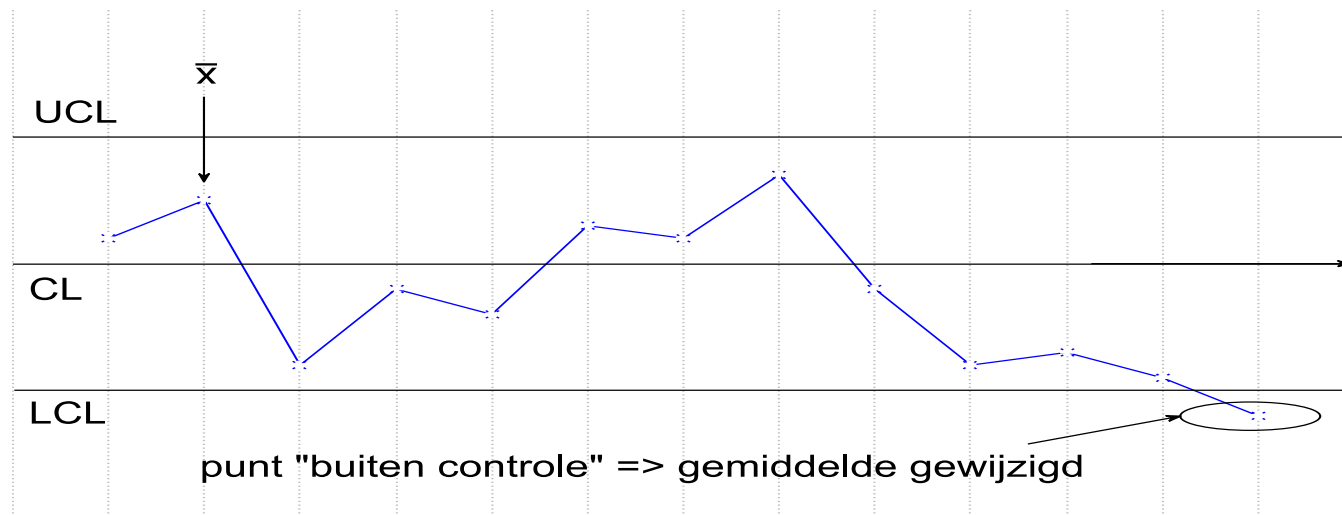
Total variation in the measurement obtained with a measurement obtained on the same master or reference value when measuring the same characteristic, over an extending time period.

Some notes on SPC

- The measurements distribution will be fixed (and quite often be normal distributed) when a process is only influenced by common causes of variation.
- Natural deteriorations are unavoidable deteriorations like wear, fatigue, ... When these occur and are not countered through e.g., good preventive maintenance, the process will become or will not be stable!
- SPC is indeed based on the normal distribution, however we cannot state that every process distribution will follow the normal law. Quite often it will be the case, but there are exceptions.
→ averages of samples follow however always normal distribution, central limit theorem

De Xbar-kaart

- Zij proceskenmerk $X \sim N(\mu_0, \sigma_0)$
- SPC volgt stabiliteit op via regelmatige controles: nagaan of μ_0 en σ_0 constant blijven (aanname: N blijft)
- controle μ_0 via zgn. $\bar{X}$ -kaart: bij elke controle (klein) sample nemen en gemiddelde berekenen; dit uitzetten in grafiek met controlelijnen (centrale limietstelling: normale verdeling van gemiddelden)



De Xbar-kaart

Distribution of individual measurements x :

Normal with mean $\mu = 1.5$ and $\sigma = 0.15$



Distribution of $\bar{x}$: Normal with mean $\mu = 1.5$ and $\sigma_{\bar{x}} = 0.0671$



Sample: $n = 5$



UCL = 1.7013

Center Line = 1.5

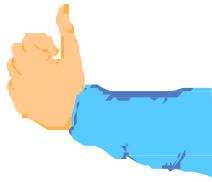
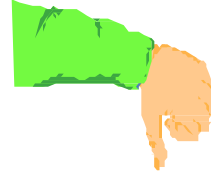
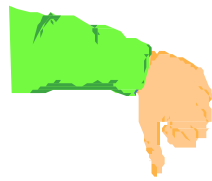
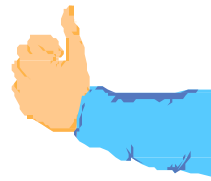
LCL = 1.2987

Review: Acting On Control Chart Information

Two Types Of Mistakes

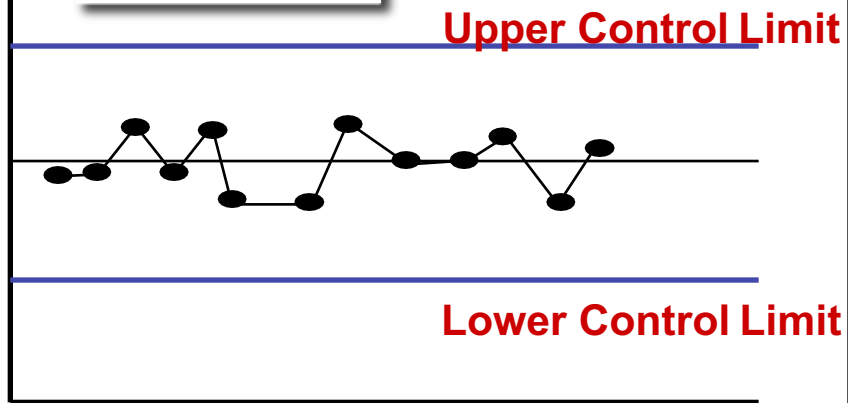
How you treat variation...

**What the
variation
really is...**

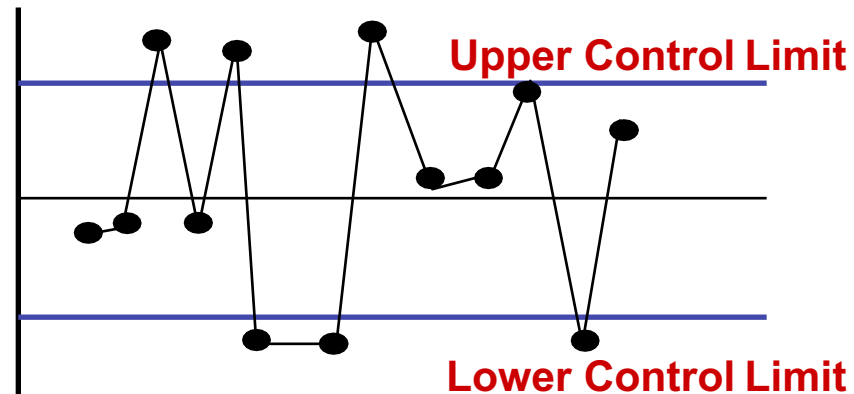
	Common Causes	Special Causes
Common Causes	Focus on fundamental process change 	Mistake 1 Tampering (increases variation) 
Special Causes	Mistake 2 Underreacting (missed prevention) 	Focus on investigating special causes 

Review: Control Limits vs. specification limits

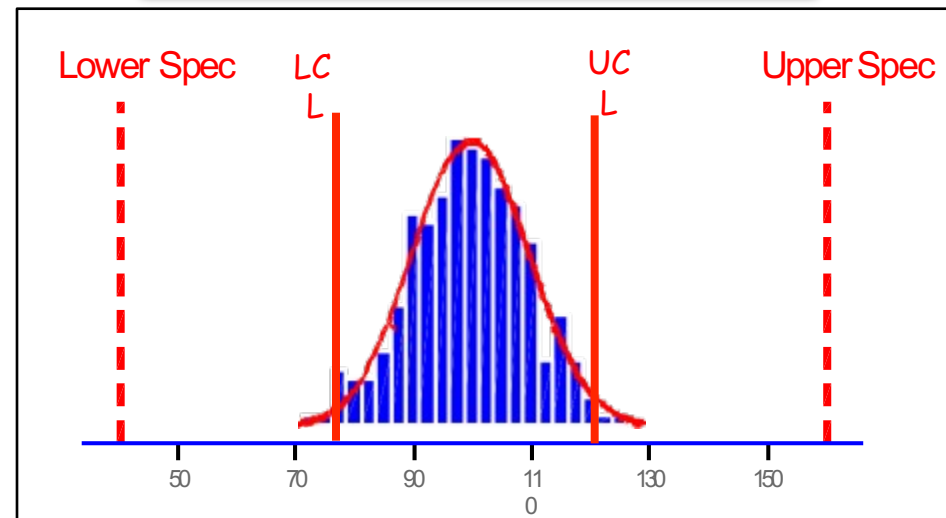
In Control



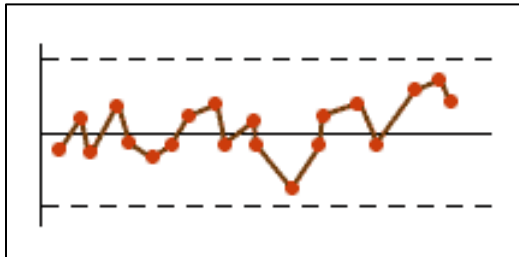
Out of Control



Capable of Meeting Specification



Control Charts Provide Decision Rules to Determine when a Process is IN CONTROL

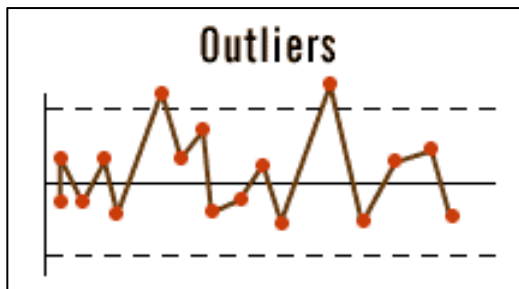


Process IN STATISTICAL CONTROL

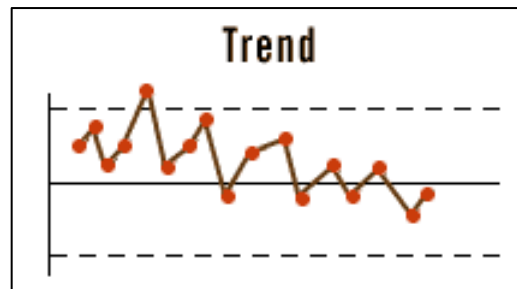
- *Random Variation around the Target*
- *All points within the control limits*

Corrective Action Needed

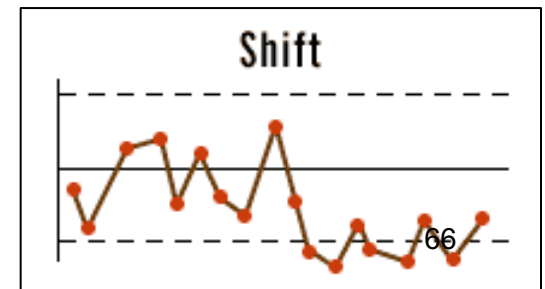
Isolated points outside control limits



Long-term trends up or down



Shifts away from the Target



Western Electric regels

■ **TABLE 5.1**

Some Sensitizing Rules for Shewhart Control Charts

Standard Action Signal:

1. One or more points outside of the control limits.
2. Two of three consecutive points outside the two-sigma warning limits but still inside the control limits.
3. Four of five consecutive points beyond the one-sigma limits.
4. A run of eight consecutive points on one side of the center line.
5. Six points in a row steadily increasing or decreasing.
6. Fifteen points in a row in zone C (both above and below the center line).
7. Fourteen points in a row alternating up and down.
8. Eight points in a row on both sides of the center line with none in zone C.
9. An unusual or nonrandom pattern in the data.
10. One or more points near a warning or control limit.

Western
Electric
Rules

Zone	Region
Zone A	Between 2σ from the centerline and the control limit (3σ)
Zone B	Between 1σ and 2σ from the centerline
Zone C	Within 1σ of the centerline

Western Electric regels

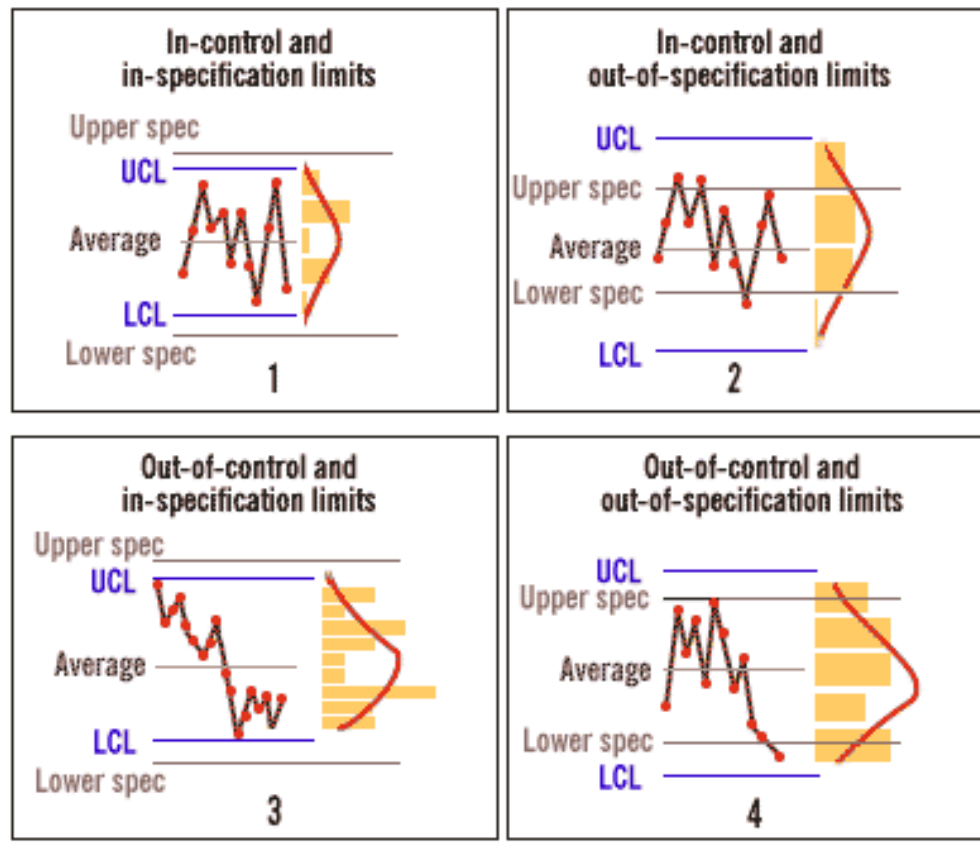
- [Western Electric rules - Wikipedia](#)

Zone	Region
Zone A	Between 2σ from the centerline and the control limit (3σ)
Zone B	Between 1σ and 2σ from the centerline
Zone C	Within 1σ of the centerline

CAUTION:

Process in Statistical Control $\neq$ Process Satisfies Customer Requirements

Four Possible States for a Process

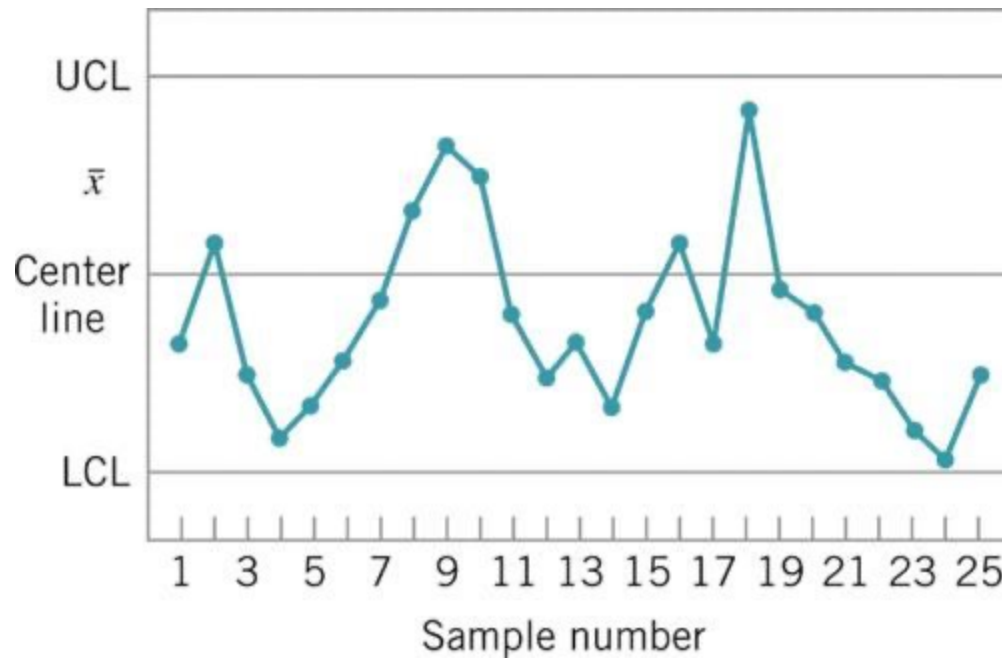


Agenda

- 13u30-15u30: deel 1 Intro & Process Capability
- 15u30-15u45: pauze
- 15u45-17u45: deel 2 Statistical Process Control (charts)
- 17u45-18u15: pauze (broodjes)
- 18u15-20u15: deel 3 Exercises

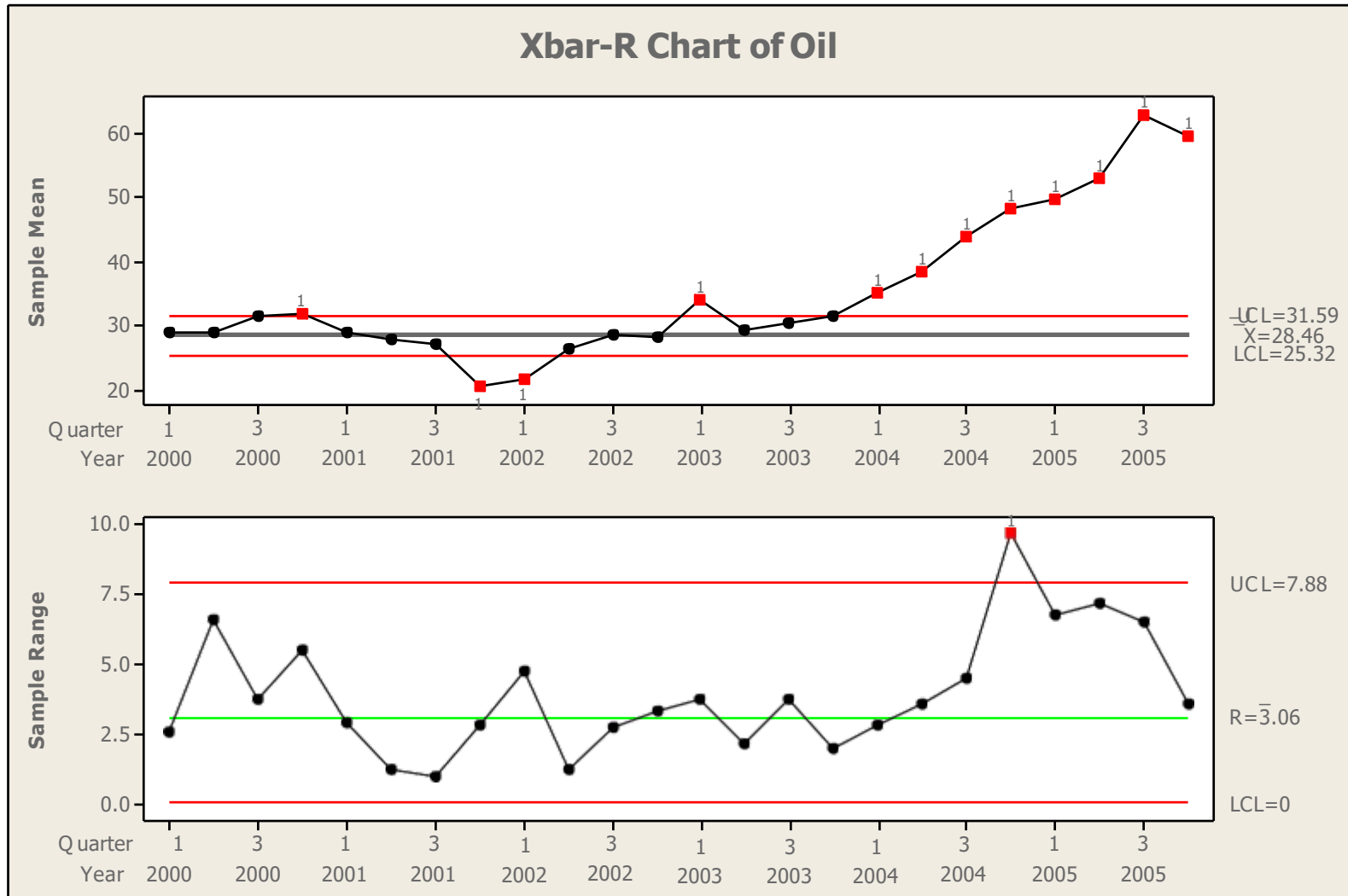
SPC: oefening 1

- Wanneer een proces statistisch onder controle is, betekent dit dan noodzakelijk dat alle of bijna alle units van het geproduceerde eindproduct binnen de specificatielimieten zullen liggen?
- Pas de Western Electric-regels toe op deze controlekaart:



X bar R Chart: 2 charts,

1 of sample main and one of sample ranges (or spread)



Xbar/R- en Xbar/s-kaart (R small sample < 10)

- Vaste sample grootte

Variabele sample grootte

$\bar{X}/R - \text{chart}$

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

$$CL = \bar{R}$$

$$UCL = D_4 \bar{R}$$

$$LCL = D_3 \bar{R}$$

$$\bar{\bar{X}} = \frac{1}{k} \sum_{i=1}^k \bar{X}_i ; \bar{R} = \frac{1}{k} \sum_{i=1}^k R_i ; \bar{s} = \frac{1}{k} \sum_{i=1}^k s_i$$

$\bar{X}/s - \text{chart}$

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_3 \bar{s}$$

$$LCL = \bar{\bar{X}} - A_3 \bar{s}$$

$$CL = \bar{s}$$

$$UCL = B_4 \bar{s}$$

$$LCL = B_3 \bar{s}$$

$\bar{X} - \text{chart}$

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + 3 \frac{\hat{\sigma}}{\sqrt{n_i}}$$

$$LCL = \bar{\bar{X}} - 3 \frac{\hat{\sigma}}{\sqrt{n_i}}$$

$$\hat{\sigma} = \frac{1}{k} \sum_{i=1}^k \hat{\sigma}_i$$

$$\hat{\sigma}_i = \frac{R_i}{d_2}$$

SPC: oefening 2

- Het nettogewicht van een bleekproduct moet opgevolgd worden dmv $\bar{X}$ en R controlekaarten, met een samplegrootte van $n=5$. Gegevens van 20 voorbereidende samples worden getoond in deze tabel.
- a) Stel de $\bar{X}$ en R controlekaarten op. Is het proces statistisch onder controle?
- b) Schat het procesgemiddelde en bepaal UCL en LCL
- c) Met $16,2 \pm 0,5$ als specs, wat kan je concluderen ivm de capabiliteit (C_p C_{pk})?

x1	x2	x3	x4	x5
15,80	16,30	16,20	16,10	16,60
16,30	15,90	15,90	16,20	16,40
16,10	16,20	16,50	16,40	16,30
16,30	16,20	15,90	16,40	16,20
16,10	16,10	16,40	16,50	16,00
16,10	15,80	16,70	16,60	16,40
16,10	16,30	16,50	16,10	16,50
16,20	16,10	16,20	16,10	16,30
16,30	16,20	16,40	16,30	16,50
16,60	16,30	16,40	16,10	16,50
16,20	16,40	15,90	16,30	16,40
15,90	16,60	16,70	16,20	16,50
16,40	16,10	16,60	16,40	16,10
16,50	16,30	16,20	16,30	16,40
16,40	16,10	16,30	16,20	16,20
16,00	16,20	16,30	16,30	16,20
16,40	16,20	16,40	16,30	16,20
16,00	16,20	16,40	16,50	16,10
16,40	16,00	16,30	16,40	16,40
16,40	16,40	16,50	16,00	15,80

Oefening 2 in Minitab

Minitab - _02 Minitab Xbar R kaart p 56 - oefening 2.mpj - [Worksheet 1 ***]

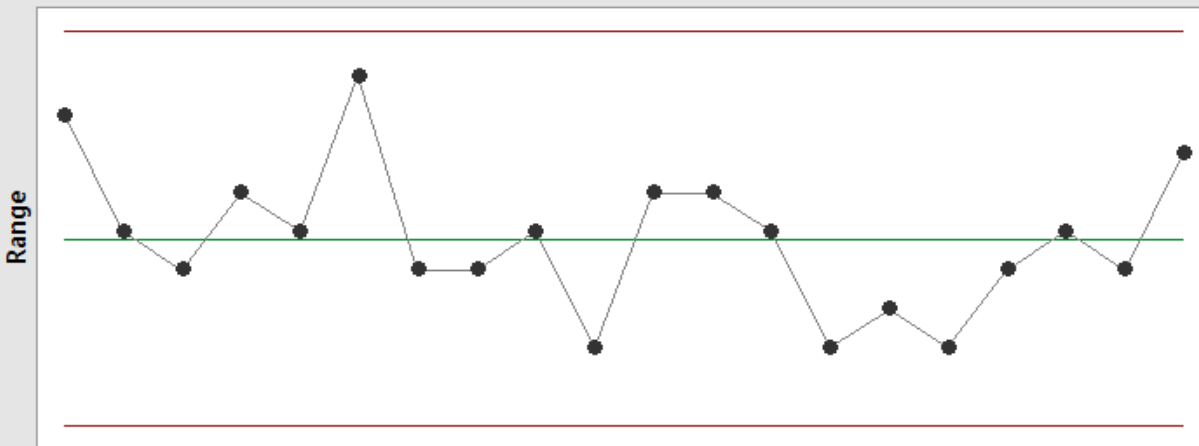
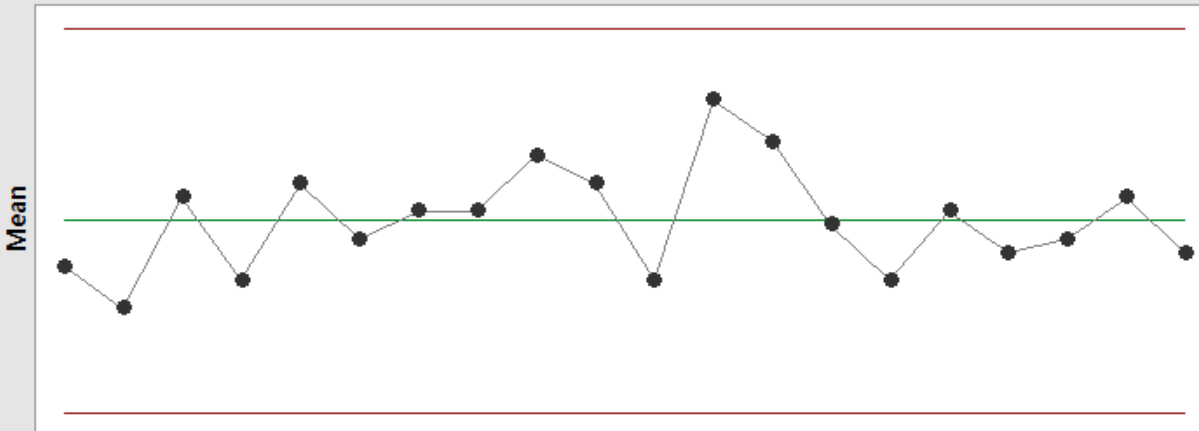
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24
1	15,8	16,3	16,2	16,1	16,6																			
2	16,3	15,9	15,9	16,2	16,4																			
3	16,1	16,2	16,5	16,4	16,3																			
4	16,3	16,2	15,9	16,5	16,0																			
5	16,1	16,1	16,4	16,6	16,4																			
6	16,1	15,8	16,7	16,1	16,5																			
7	16,2	16,3	16,5	16,1	16,3																			
8	16,3	16,1	16,2	16,3	16,5																			
9	16,6	16,2	16,4	16,1	16,5																			
10	16,2	16,3	16,4	16,3	16,4																			
11	15,9	16,4	15,9	16,2	16,5																			
12	16,4	16,6	16,7	16,4	16,1																			
13	16,5	16,1	16,6	16,3	16,4																			
14	16,4	16,3	16,2	16,2	16,2																			
15	16,0	16,1	16,3	16,3	16,2																			
16	16,4	16,2	16,3	16,3	16,2																			
17	16,0	16,2	16,4	16,3	16,2																			
18	16,0	16,2	16,4	16,5	16,1																			
19	16,4	16,0	16,3	16,4	16,4																			
20	16,4	16,4	16,5	16,0	15,8																			
21																								
22																								
23																								
24																								
25																								
26																								
27																								
28																								
29																								
30																								

Current Worksheet: Worksheet 1

Xbar-R Chart of C1; ...; C5 Report Card

Check	Status	Description
Stability		The process mean and variation are stable. No subgroups are out of control on either chart.
Amount of Data		You do not need to be concerned about the precision of your control limits because 100 or more data points are included in the calculations.
Correlated Data		If the data are correlated, you may see an increased number of false alarms. Because less than 2% of the subgroups are outside the control limits on the Xbar chart, the correlation test is not needed.
Alternative Charts		This chart is intended to monitor process control. If your primary objective is to explore your data or compare your process before and after a change, use the Graphical Analysis Control Charts or the Before/After Control Charts.

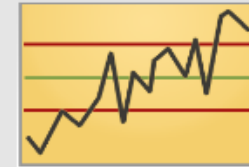
Xbar-R Chart of C1; ...; C5 Stability Report



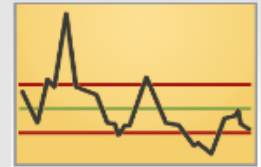
Assess the stability of the mean and variation of your process and look for patterns that can help you distinguish between common and special causes. Typically, a process that exhibits only common causes has a constant mean and constant variability. However, global trends or cyclical patterns may also be common causes. Other patterns, such as shifts and drifts, may be special causes.

Look for these patterns:

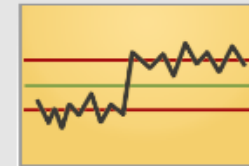
Global Trend



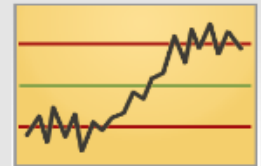
Cyclical



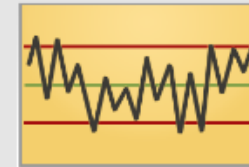
Shifts



Drifts



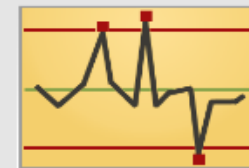
Oscillation



Mixture



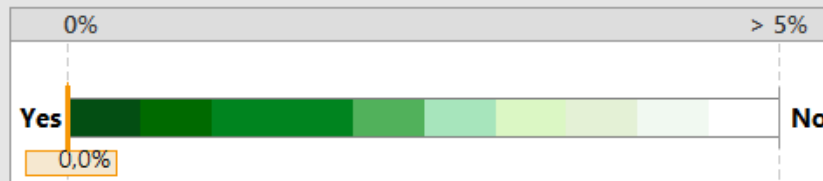
Excessive Out of Control



Xbar-R Chart of C1; ...; C5 Summary Report

Is the process mean stable?

Evaluate the % of out-of-control subgroups.

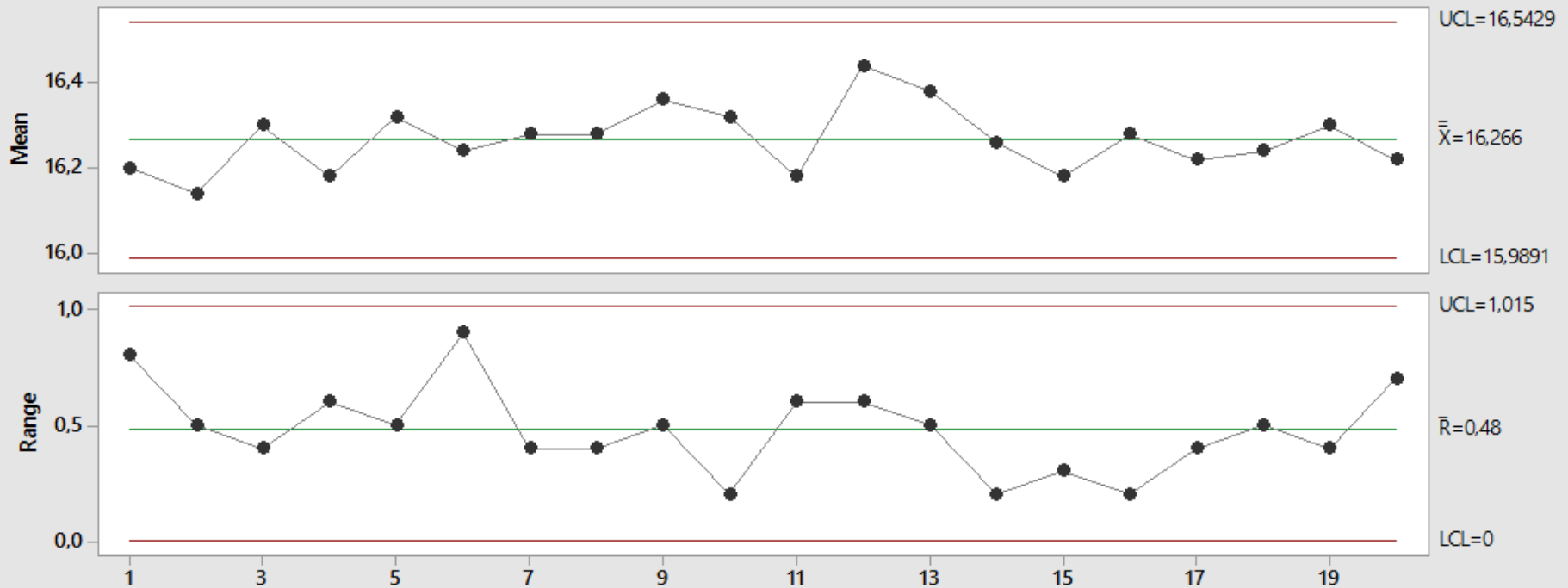


Comments

The process mean is stable. No subgroups are out of control on the Xbar chart.

Xbar and R Charts

Investigate any out-of-control subgroups.



Subgroups: 20

Mean: 16,266

StDev(within): 0,20636

StDev(overall): 0,20212

78

Control limits are estimated using the StDev(within).

Capability Analysis for C1; ... Report Card

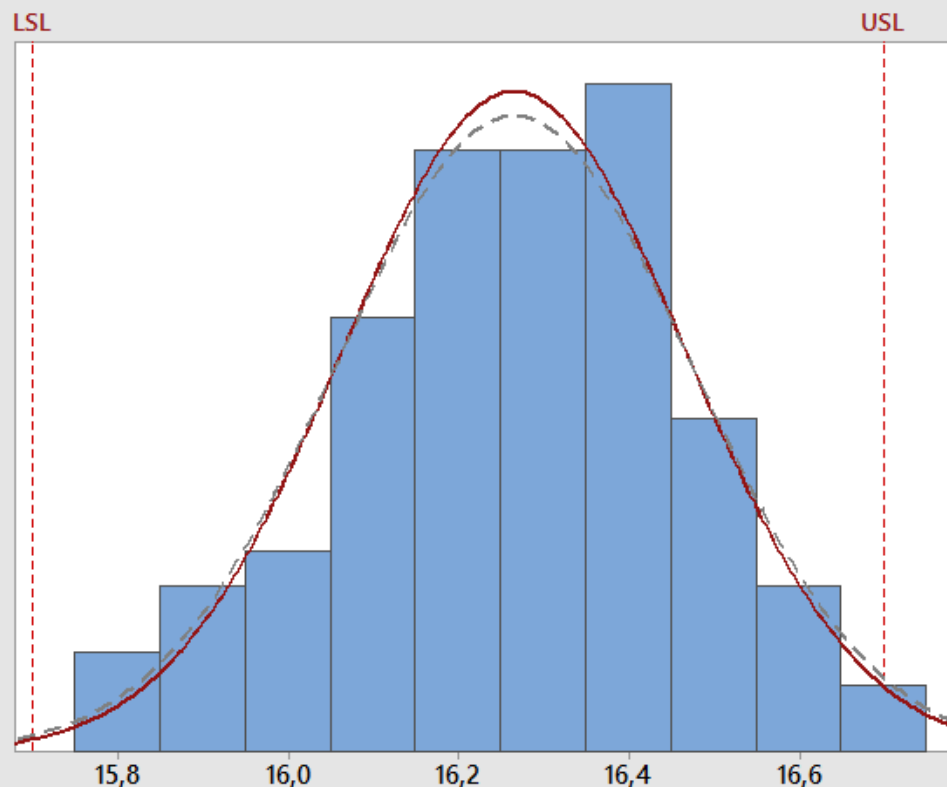
Check	Status	Description
Stability		The process mean and variation are stable. No points are out of control.
Number of Subgroups		You only have 20 subgroups. For a capability analysis, it is generally recommended that you collect at least 25 subgroups over a long enough period of time to capture the different sources of process variation.
Normality		Your data failed the normality test. A Box-Cox transformation will not correct the problem. Get help to determine next steps because the capability estimates may be inaccurate.
Amount of Data		The total number of observations is 100 or more. The capability estimates should be reasonably precise.

Capability Analysis for C1; ...

Process Performance Report

Capability Histogram

Are the data inside the limits?



— Actual (overall) capability is what the customer experiences.

- - - Potential (within) capability is what could be achieved if process shifts and drifts were eliminated.

Process Characterization

Total N	100
Subgroup size	5
Mean	16,266
Standard deviation (overall)	0,20212
Standard deviation (within)	0,20982

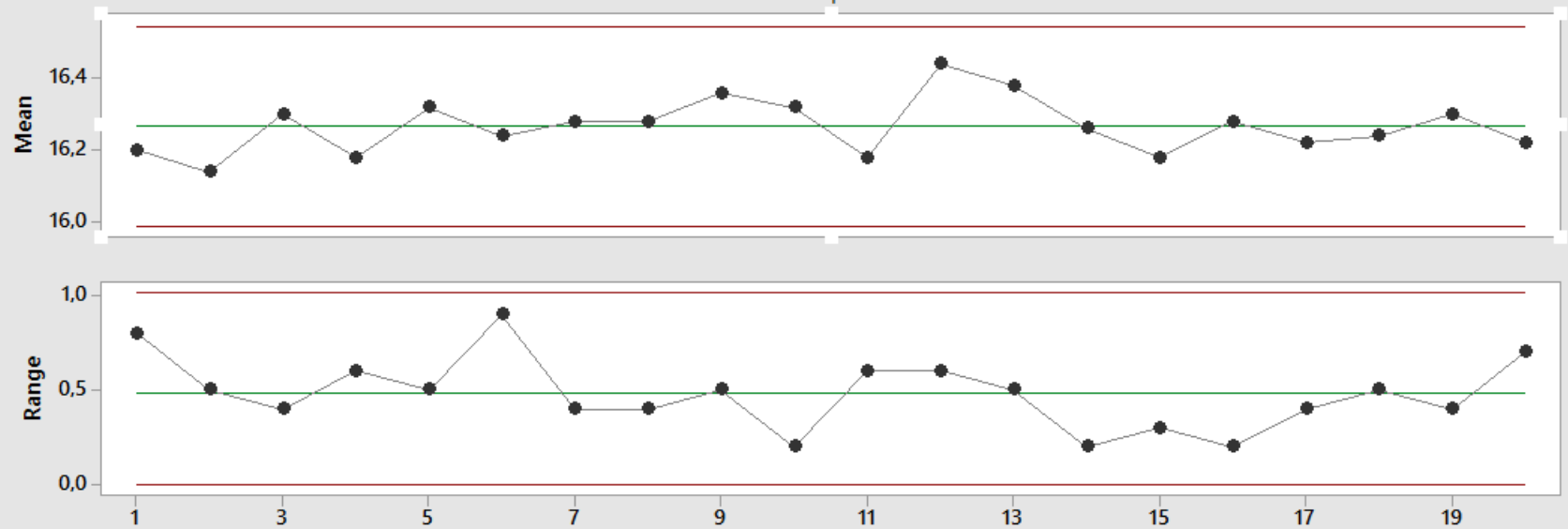
Capability Statistics

Actual (overall)	
Pp	0,82
Ppk	0,72
Z.Bench	2,09
% Out of spec (observed)	0,00
% Out of spec (expected)	1,84
PPM (DPMO) (observed)	0
PPM (DPMO) (expected)	18440
Potential (within)	
Cp	0,79
Cpk	0,69
Z.Bench	2,00
% Out of spec (expected)	2,28
PPM (DPMO) (expected)	22792

Capability Analysis for C1; ... Diagnostic Report

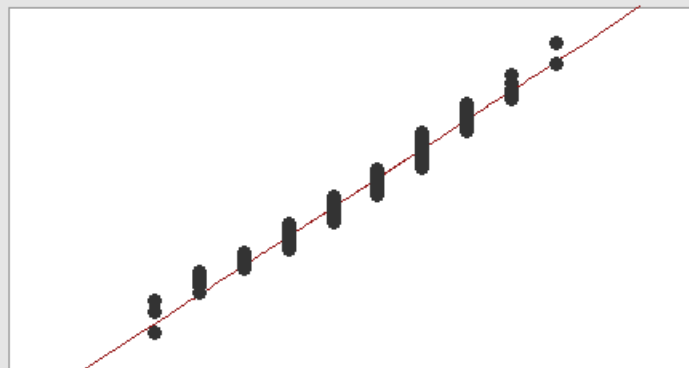
Xbar-R Chart

Confirm that the process is stable.



Normality Plot

The points should be close to the line.

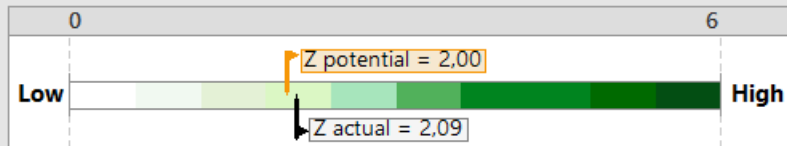


Normality Test (Anderson-Darling)

Results	Fail
P-value	< 0,005

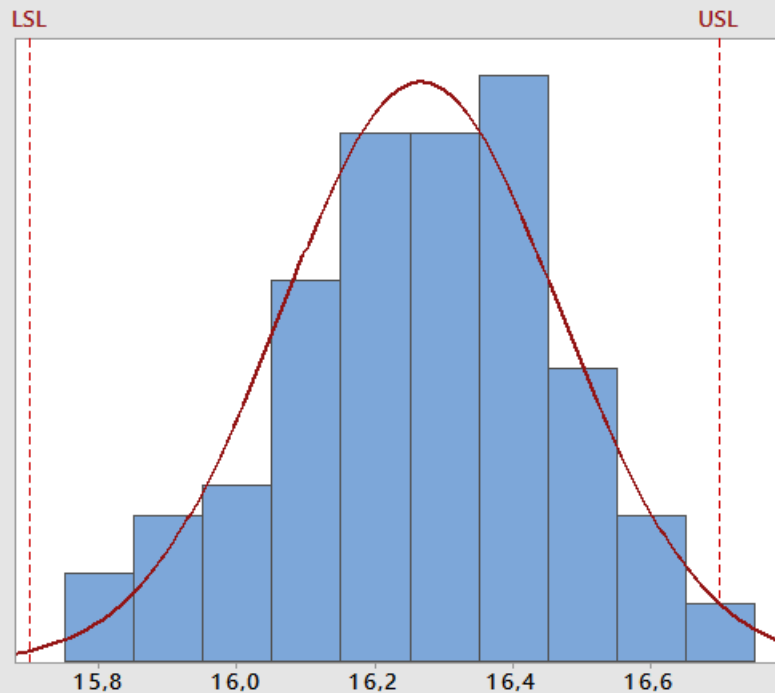
Capability Analysis for C1; ... Summary Report

How capable is the process?



Actual (Overall) Capability

Are the data inside the limits?



Customer Requirements

Upper spec	16,7
Target	*
Lower spec	15,7

Process Characterization

Mean	16,266
Standard deviation (overall)	0,20212

Actual (overall) capability

Pp	0,82
Ppk	0,72
Z.Bench	2,09

% Out of spec	1,84
PPM (DPMO)	18440

Comments

- The defect rate is 1,84%, which estimates the percentage of parts from the process that are outside the spec limits.

Actual (overall) capability is what the customer experiences.

Potential (within) capability is what could be achieved if process shifts and drifts were eliminated.

SPC: oefening 3

- De gegeven data zijn $\bar{X}$ en R waarden voor 24 samples van grootte $n=5$, genomen uit een proces dat lagers produceert. De metingen werden uitgevoerd op de binnendiameter van de lager, met enkel de laatste 3 decimalen genoteerd. (bv. 34,5 moet 0,50345 zijn).
- Stel de $\bar{X}$ en R controlekaarten op. Is het proces statistisch onder controle?
- Indien nodig, herzie de testcontrolelimieten.

SPC: oefening 4

- Neem de $\bar{x}$ en R kaarten uit de vorige oefening, met $n=5$.
- a) Stel dat je dit proceskenmerk verder zou willen monitoren met $\bar{X}$ en R kaarten gebaseerd op een samplegrootte van $n=3$. Welke limieten zou je dan gebruiken?
- b) Wat zouden de limieten zijn indien $n=8$?
- c) Wat zou de impact van deze beslissingen zijn op de mogelijkheid om een verschuiving van $2 \cdot \sigma$ van het gemiddelde te ontdekken?

SPC: oefening 5

- Samples van grootte $n=5$ worden elk uur uit een productieproces genomen. De kwaliteitskarakteristiek is normaal verdeeld. $\bar{X}$ en R worden voor elke sample gemeten. Na 25 samples hebben we volgende gegevens:

$$n = 5; \quad \sum_{i=1}^{25} \bar{x}_i = 662.50; \quad \sum_{i=1}^{25} R_i = 9.00; \quad m = 25 \text{ samples}$$

- Veronderstel dat beide kaarten statistisch onder controle zijn. Indien de specificaties $26,40 \pm 0,50$ bedragen, wat is dan de procescapabiliteit? Wat is het percentage aan niet-correcte eenheden dat we mogen verwachten?
- Wat zou dit percentage bedragen indien het procesgemiddelde 26,40 zou zijn?
 - (hint: gebruik de cumul. standaardnormale verdeling)

SPC: oefening 6

- Een kritische dimensie van een gefreesd stuk heeft als specificaties 100 ± 10 . Analyse van de controlekaart toont aan dat het proces onder controle is met $\bar{X} = 104$ en $\bar{R} = 9,30$. De samplegrootte is $n=5$.
 $s = \bar{R} / d_2 = 9,3/2,326 \approx 4$
- Indien we aannemen dat de karakteristiek normaal verdeeld is, kunnen we dan het gemiddelde zo aanpassen (door de freespositie aan te passen) dat alle output voldoet aan de specificaties?
Neen, $3 \sigma = 12$: buiten specs
- Wat is de huidige capabiliteit van het proces?
 $C_p = 20/24 = \mathbf{0,83}$
 $C_{pk} = \min [(104-90)/(3.4) ; (110-104)/(3.4)] = \min [1,17 ; \mathbf{0,5}]$

- Voorbeelden OSPC AMGent

Ondersteuning voor bewaken productkwaliteit

OSPC controle door productie – bewaking meetmiddelen

OCAP

